

Contents: *Frequency Response Analysis*

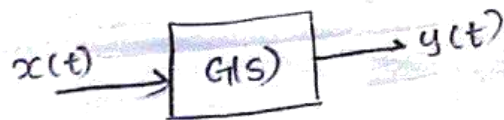
- *Bode plot*
- *Polar Plot*
- *Nyquist Plot*



Control Systems

UNIT-4 FREQUENCY RESPONSE ANALYSIS

The frequency response is the steady state response of a system when the input to the system is sinusoidal signals.



$$x(t) = X \sin \omega t, \quad y(t) = Y \sin(\omega t + \phi)$$

$y(t)$ is also a sinusoidal signal as $x(t)$ of same frequency but with different magnitude and phase angle.

The frequency response of the system is normally obtained by varying the frequency of the input signal by keeping the magnitude of the input signal at a constant value.

In the system Transfer function, s is replaced by $j\omega$, then the resulting transfer function is called sinusoidal Transfer function.

$$\text{Open Loop Transfer function } G(j\omega) = |G(j\omega)| \angle G(j\omega)$$

$$\text{Loop Transfer function } G(j\omega)H(j\omega) = |G(j\omega)H(j\omega)| \angle G(j\omega)H(j\omega)$$

$$\text{closed loop Transfer function } \frac{C(j\omega)}{R(j\omega)} = |M(j\omega)| \angle M(j\omega)$$

Advantages of Frequency Response Analysis :

1. Absolute and Relative stability can be estimated from the knowledge of their open loop frequency

Response.

2. Transfer function of complicated systems can be determined experimentally by frequency Response test.
3. Design and parameter adjustment of the open loop Transfer function can be carried out more easily in frequency domain.
4. The effects of noise disturbances and parameter variations are relatively easy to visualise and incorporate corrective measures.
5. Frequency Response Analysis and designs can be extended to certain non-linear control systems.

Frequency Domain specifications :

1. Resonant Peak (M_r) :
2. Resonant Frequency (ω_r)
3. Bandwidth
4. cutoff rate
5. Gain Margin
6. Phase margin

1. Resonant Peak (M_r) :

The maximum value of magnitude of closed loop transfer function is called Resonant peak.

A large Resonant peak corresponds to large overshoot in transient response.

Derivation :

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Relace s by $j\omega$ in above eqn.

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$m(j\omega) = \frac{C(j\omega)}{R(j\omega)} = \frac{1}{1 - \left(\frac{\omega}{\omega_n}\right)^2 + j \frac{2\xi\omega}{\omega_n}}$$

$$\Rightarrow |m(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[\frac{2\xi\omega}{\omega_n}\right]^2}} \quad \text{— magnitude}$$

$$-\tan^{-1}\left[\frac{2\xi\omega/\omega_n}{1 - \left(\frac{\omega}{\omega_n}\right)^2}\right] = \angle m(j\omega) \quad \text{— phase.}$$

The Resonant peak is the maximum value of m .
The condition for maximum value of m can be obtained by differentiating the magnitude of $m(j\omega)$ i.e., $|m(j\omega)|$ w.r.t ω

$$\begin{aligned} \frac{d|m(j\omega)|}{d\omega} &= \frac{d}{d\omega} \left[\left[1 - \left(\frac{\omega}{\omega_n}\right)^2 \right]^2 + \left[\frac{2\xi\omega}{\omega_n} \right]^2 \right]^{-1/2} \\ &= -\frac{1}{2} \left[\left[1 - \left(\frac{\omega}{\omega_n}\right)^2 \right]^2 + \left[\frac{2\xi\omega}{\omega_n} \right]^2 \right]^{-3/2} \left[2 \left(1 - \left(\frac{\omega}{\omega_n}\right)^2 \right) \left(-\frac{2\omega}{\omega_n} \right) + \frac{(2\xi)^2 \cdot 2\omega}{\omega_n} \right] \end{aligned}$$

$$\frac{d|m(j\omega)|}{d\omega} = 0$$

to replace ω with ω_r

$$\Rightarrow -\frac{1}{2 \sqrt{\left[1 - \left(\frac{\omega_r}{\omega_n}\right)^2 \right]^2 + \left[\frac{2\xi\omega_r}{\omega_n} \right]^2}} \left[2 \left(1 - \left(\frac{\omega_r}{\omega_n}\right)^2 \right) \left(-\frac{2\omega_r}{\omega_n} \right) + \frac{(2\xi)^2 (2\omega_r)}{\omega_n} \right] = 0$$

$$2 \left(1 - \left(\frac{\omega r}{\omega_n} \right)^2 \right) \left(-\frac{2\omega r}{\omega_n} \right) = \left[\frac{(2\xi)^2 (2\omega r)}{\omega_n} \right]$$

$$\left[2 \sqrt{1 - \left(\frac{\omega r}{\omega_n} \right)^2} \right]$$

$$\therefore \left[2 \left(1 - \left(\frac{\omega r}{\omega_n} \right)^2 \right) \left(-\frac{2\omega r}{\omega_n} \right) + \frac{(2\xi)^2 \cdot 2\omega}{\omega_n} \right] = 0$$

$$\Rightarrow 2 \left(1 - \left(\frac{\omega r}{\omega_n} \right)^2 \right) \left(-\frac{2\omega r}{\omega_n} \right) = -\frac{8\xi^2 \omega r}{\omega_n}$$

$$\Rightarrow \left[2 - \frac{2\omega r^2}{\omega_n^2} \right] \left[-2\omega r \right] = -8\xi^2 \omega r$$

$$\Rightarrow 2 - \frac{2\omega r^2}{\omega_n^2} = +4\xi^2$$

$$\Rightarrow 2 + 4\xi^2 = \frac{2\omega r^2}{\omega_n^2}$$

$$\Rightarrow 1 + 2\xi^2 = \frac{\omega r^2}{\omega_n^2}$$

$$\Rightarrow \omega r^2 = \omega_n^2 (1 + 2\xi^2)$$

$$\Rightarrow \boxed{\omega r = \omega_n \sqrt{1 + 2\xi^2}}$$

substitute ωr value in m. 1

$$|M(\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega_n \sqrt{1 + 2\xi^2}}{\omega_n} \right)^2 \right]^2 + \left[\frac{2\xi \omega_n \sqrt{1 + 2\xi^2}}{\omega_n} \right]^2}}$$

$$= \frac{1}{\sqrt{\left[1 - (\sqrt{1 + 2\xi^2})^2 \right]^2 + (2\xi \sqrt{1 + 2\xi^2})^2}}$$

$$= \frac{1}{\sqrt{(2\xi^2)^2 + (1 - 2\xi^2)4\xi^2}}$$

$$= \frac{1}{\sqrt{4\xi^4 + 4\xi^2 - 8\xi^4}}$$

$$= \frac{1}{\sqrt{4\xi^2 - 4\xi^4}}$$

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}}$$

$$M_r = \frac{1}{2\xi\sqrt{1-\xi^2}} \quad \text{for } \xi < 0.707.$$

* Bandwidth (ω):

At $\omega = \omega_b$, the magnitude of closed loop system is $1/\sqrt{2}$.

$$\Rightarrow \frac{1}{\sqrt{\left(1 - \left(\frac{\omega_b}{\omega_n}\right)^2\right)^2 + \left(2\xi \frac{\omega_b}{\omega_n}\right)^2}} = \frac{1}{\sqrt{2}}$$

Squaring on both sides

$$\Rightarrow \left(1 - \frac{\omega_b^2}{\omega_n^2}\right)^2 + \left(2\xi \frac{\omega_b}{\omega_n}\right)^2 = 2.$$

$$\Rightarrow \frac{(\omega_n^2 - \omega_b^2)^2}{\omega_n^4} + \frac{(2\xi \omega_b)^2}{\omega_n^2} = 2$$

$$\Rightarrow \omega_n^4 + \omega_b^4 - 2\omega_n^2\omega_b^2 + 4\xi^2\omega_b^2\omega_n^2 = 2\omega_n^4$$

$$\Rightarrow \omega_b^4 - 2\omega_n^2\omega_b^2 + 4\xi^2\omega_b^2\omega_n^2 = \omega_n^4$$

$$\Rightarrow \omega_b^2 \left[\omega_b^2 - 2\omega_n^2(1 - 2\xi^2) \right] = \omega_n^4$$

$$\left(\frac{\omega_b}{\omega_n}\right)^4 + 4\xi^2\left(\frac{\omega_b}{\omega_n}\right)^2 - 2\left(\frac{\omega_b}{\omega_n}\right)^2 = 0$$

$$\left(\frac{\omega_b}{\omega_n}\right)^4 + (4\xi^2 - 2)\left(\frac{\omega_b}{\omega_n}\right)^2 - 1 = 0$$

$$\left(\frac{\omega_b}{\omega_n}\right)^2 = \frac{-(4\xi^2 - 2) \pm \sqrt{(4\xi^2 - 2)^2 + 4}}{2}$$

$$\left(\frac{\omega_b}{\omega_n}\right)^2 = \frac{-(4\xi^2 - 2) \pm \sqrt{16\xi^4 + 4 - 16\xi^2 + 4}}{2}$$

$$\left(\frac{\omega_b}{\omega_n}\right)^2 = (1 - 2\xi^2) \pm \sqrt{4\xi^4 + 2 - 4\xi^2}$$

[∵ BW must be real quantity]

$$\Rightarrow \frac{\omega_b}{\omega_n} = \left[(1 - 2\xi^2) + \sqrt{4\xi^4 + 2 - 4\xi^2} \right]^{1/2}$$

$$\Rightarrow \omega_b = \omega_n \left[(1 - 2\xi^2) + \sqrt{4\xi^4 + 2 - 4\xi^2} \right]^{1/2}$$

* Gain margin :

$$\text{Gain margin } K_g = \frac{1}{|G(j\omega_{pc})|}$$

$$\text{Gain margin in dB} = K_g \text{ in dB} = 20 \log(K_g)$$

$$= 20 \log \left[\frac{1}{|G(j\omega_{pc})|} \right]$$

$$= -20 \log [|G(j\omega_{pc})|]$$

Gain margin indicates the amount by which the gain of the system can be increased without effecting the stability of the system.

* Phase margin :

Phase margin is obtained by adding 180° to the phase angle ϕ of the open loop transfer

function at gain cross over frequency.

$$\text{O.L.T.F } G(s) = \frac{\omega_n^2}{s(s + 2\xi\omega_n)}$$

Replace s by $j\omega$

$$\Rightarrow G(j\omega) = \frac{\omega_n^2}{- \omega^2 + 2\xi\omega_n j}$$

$$\Rightarrow G(j\omega) = \frac{1}{-\left(\frac{\omega}{\omega_n}\right)^2 + 2\xi j \left(\frac{\omega}{\omega_n}\right)}$$

$$|G(j\omega)| = \frac{1}{\sqrt{\left(-\frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\xi \frac{\omega}{\omega_n}\right)^2}}$$

$$\angle G(j\omega) = -\tan^{-1} \left[\frac{2\xi \omega / \omega_n}{-\left(\frac{\omega}{\omega_n}\right)^2} \right]$$

$$= \tan^{-1} \left[\frac{2\xi \omega}{\omega_n} \right]$$

At gain cross over frequency (ω_{gc}) the magnitude of $G(j\omega) = 1$

$$\Rightarrow \frac{1}{\sqrt{\left(-\frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\xi \frac{\omega}{\omega_n}\right)^2}} = 1$$

Replace ω by ω_{gc}

$$\Rightarrow \left(-\frac{\omega_{gc}^2}{\omega_n^2}\right)^2 + \left(2\xi \frac{\omega_{gc}}{\omega_n}\right)^2 = 1$$

$$\Rightarrow \frac{(\omega_{gc})^4}{\omega_n^4} + 4\xi^2 \frac{\omega_{gc}^2}{\omega_n^2} = 1$$

$$\Rightarrow \left[\left(\frac{\omega_{gc}}{\omega_n} \right)^2 \right]^2 + 4\xi^2 \frac{\omega_{gc}^2}{\omega_n^2} - 1 = 0$$

Find roots for $\left(\frac{\omega_{gc}}{\omega_n} \right)^2$

$$\Rightarrow \left(\frac{\omega_{gc}}{\omega_n} \right)^2 = \frac{-4\xi^2 \pm \sqrt{16\xi^4 + 4}}{2}$$

$$\left(\frac{\omega_{gc}}{\omega_n} \right)^2 = -2\xi^2 \pm \sqrt{4\xi^4 + 1}$$

$$\Rightarrow \left(\frac{\omega_{gc}}{\omega_n} \right) = \left[-2\xi^2 \pm \sqrt{4\xi^4 + 1} \right]^{1/2}$$

$$\Rightarrow \omega_{gc} = \omega_n \left[-2\xi^2 \pm \sqrt{4\xi^4 + 1} \right]^{1/2}$$

Phase margin

$$PM = 180^\circ + \angle G(j\omega) \text{ at } \omega_{gc}$$

$$PM = 180^\circ + \tan^{-1} \frac{2\xi\omega_n}{\omega_{gc}}$$

$$PM = 180^\circ + \tan^{-1} \frac{\omega_n}{\omega_n \left[-2\xi^2 \pm \sqrt{4\xi^4 + 1} \right]^{1/2}}$$

* Correlation between time and Frequency Response :

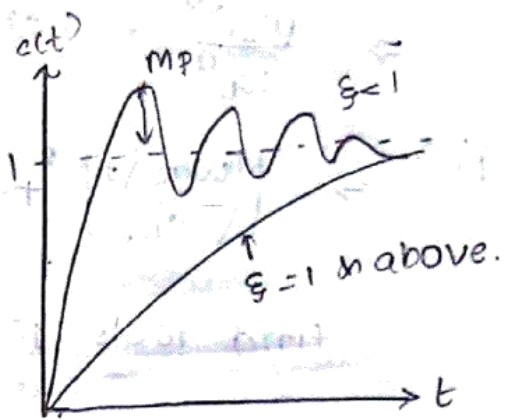
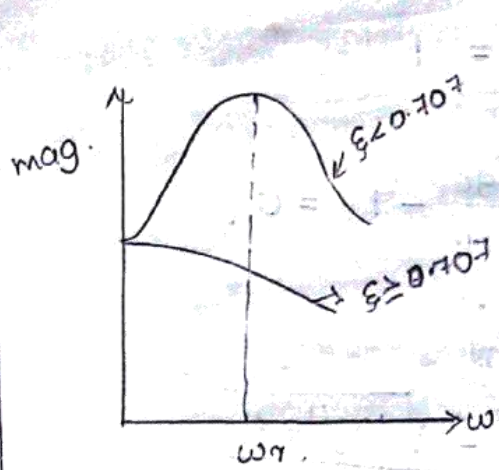
$$1. \omega_n = \omega_n \sqrt{1 - 2\xi^2}$$

$$2. M_n = \frac{1}{2\xi \sqrt{1 - \xi^2}}$$

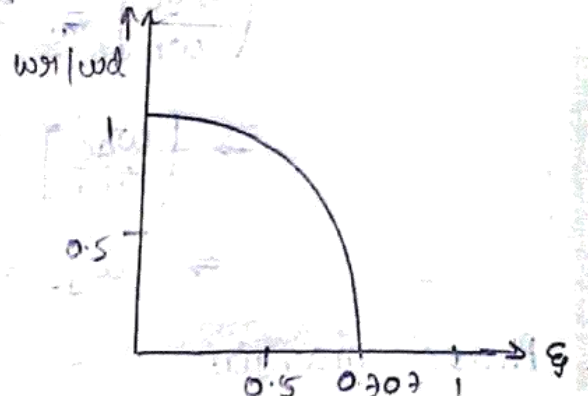
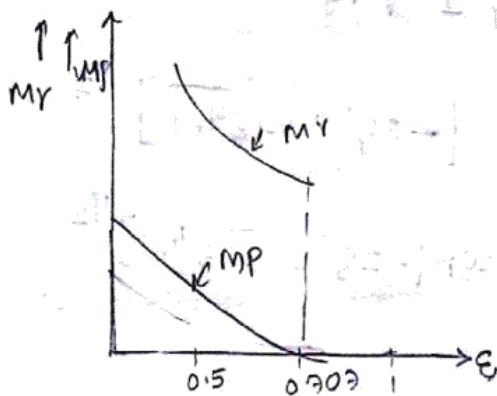
$$\omega_d = \omega_n \left[1 - 2\xi^2 + \sqrt{2 - 4\xi^2 + 4\xi^4} \right]^{1/2}$$

$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

$$m_p = e^{-\xi\pi/\sqrt{1-\xi^2}}$$



with respect to ξ



As $\xi \rightarrow 0$, ω_r and $\omega_d \rightarrow \omega_n$ and $M_p \rightarrow \infty$, M_p achieves 100% overshoot and $\omega_d \rightarrow \omega_n$

- When ξ is small, ω_d is large and hence rise time is small. When ξ is small, ω_r is large and hence system is more fast in response.

Hence ω_r indicates speed of response.

* Frequency Response Plots :

Frequency Response Analysis of control system can be carried either analytically or graphically. Various graphical techniques are available for frequency response Analysis.

1. Bode plot
2. Polar plot. or Nyquist plot.

3. Nicholas plot
4. M and N circles
5. Nicholas chart.

Limitations of Frequency Response plots:

* 1. BODE PLOT :-

Bode plot is a frequency response plot of the transfer function of the system.

- It consists of two graphs.
 - one is a plot of magnitude of sinusoidal transfer function vs $\log \omega$.
 - other is a plot of phase angle of sinusoidal transfer function vs $\log \omega$.
- The standard representation of logarithmic magnitude of open loop transfer function & $G(j\omega)H(j\omega)$ is $20 \log_{10} |G(j\omega)H(j\omega)|$.
- The main advantage of Bode plot is that multiplication of magnitudes can be converted into additions.

→ consider open loop transfer function

$$G(s)H(s) = \frac{k(1+sT_1)}{s(1+sT_2)(1+sT_3)}$$

1. Replace s by $j\omega$

$$G(j\omega)H(j\omega) = \frac{k(1+j\omega T_1)}{j\omega(1+j\omega T_2)(1+j\omega T_3)}$$

$$= \frac{k \angle 0^\circ \sqrt{1+\omega^2 T_1^2} \angle \tan^{-1}(T_1 \omega)}{\omega \angle 90^\circ \sqrt{1+\omega^2 T_2^2} \angle \tan^{-1}(T_2 \omega) \cdot \sqrt{1+\omega^2 T_3^2} \angle \tan^{-1}(T_3 \omega)}$$

$$|G(j\omega)| = \frac{K \sqrt{1+\omega^2 T_1^2}}{\omega \sqrt{1+\omega^2 T_2^2} \cdot \sqrt{1+\omega^2 T_3^2}}$$

$$\angle G(j\omega) = \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2) - \tan^{-1}(\omega T_3) - 90^\circ$$

$$|G(j\omega)| \text{ in dB} = 20 \log \left[\frac{K \sqrt{1+\omega^2 T_1^2}}{\omega \sqrt{1+\omega^2 T_2^2} \cdot \sqrt{1+\omega^2 T_3^2}} \right]$$

$$= 20 \log \left[\frac{K}{\omega} \cdot \sqrt{1+\omega^2 T_1^2} \cdot \frac{1}{\sqrt{1+\omega^2 T_2^2}} \cdot \frac{1}{\sqrt{1+\omega^2 T_3^2}} \right]$$

$$= 20 \log \left[\frac{K}{\omega} \right] + 20 \log [\sqrt{1+\omega^2 T_1^2}] -$$

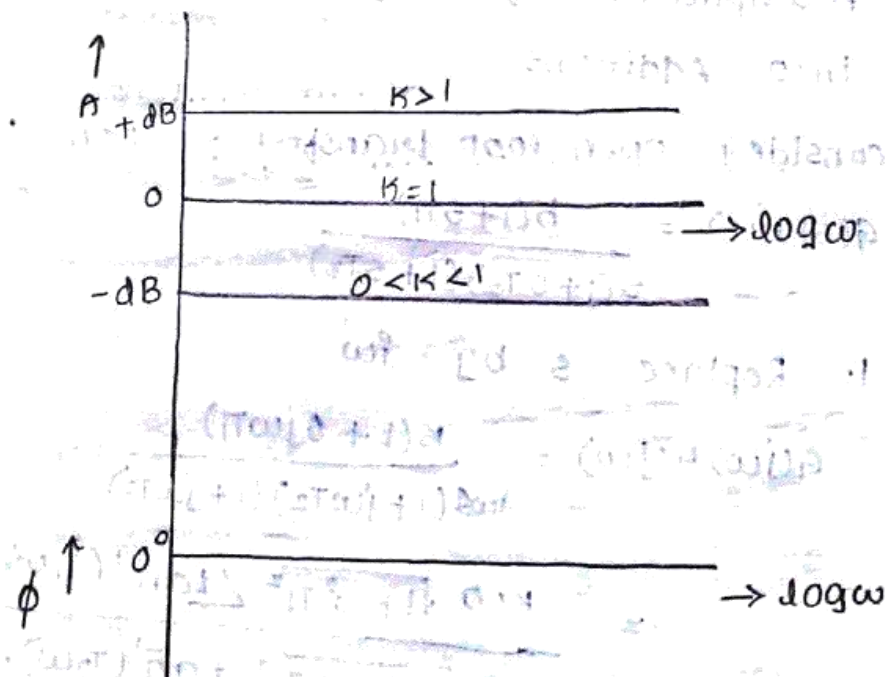
$$20 \log [\sqrt{1+\omega^2 T_2^2}] - 20 \log [\sqrt{1+\omega^2 T_3^2}]$$

∴ constant gain, K :

$$\text{Let } G(s) = K \Rightarrow G(j\omega) = K = K \angle 0^\circ$$

$$A = |G(j\omega)| \text{ in dB} = 20 \log (K)$$

$$\phi = \angle G(j\omega) = 0^\circ$$



When $K > 1$, $20 \log K$ is positive

or $0 < K < 1$, $20 \log K$ is negative

$K = 1$, $20 \log K$ is zero.

2. Integral factor:

$$\text{Let } G(s) = \frac{k}{s}$$

$$\Rightarrow G(j\omega) = \frac{k}{j\omega}$$

$$|G(j\omega)| = \frac{k}{\omega} \angle -90^\circ$$

$$A = |G(j\omega)| \text{ in dB} = 20 \log \left(\frac{k}{\omega} \right)$$

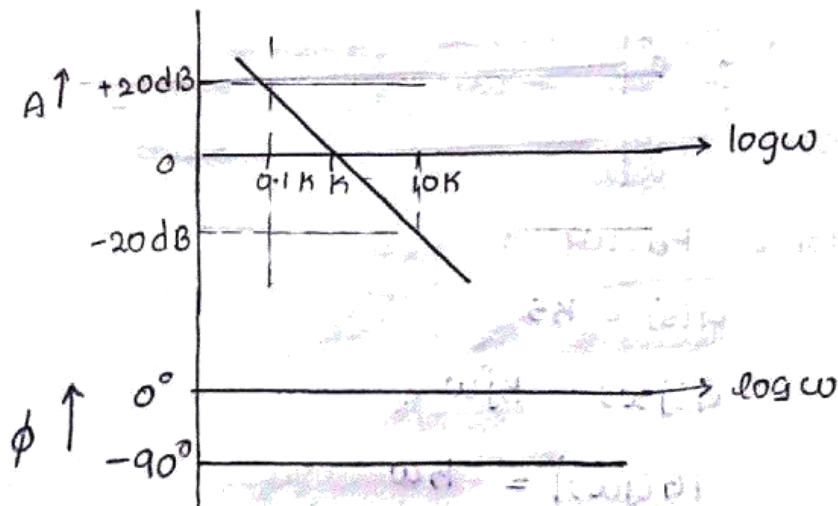
$$\phi = \angle G(j\omega) = -90^\circ$$

When $\omega = 0.1K$

$$\Rightarrow A = 20 \log \left[\frac{k}{0.1K} \right] = 20 \text{ dB}$$

When $\omega = K$, $A = 20 \log \left[\frac{k}{K} \right] = 0 \text{ dB}$

When $\omega = 10K$, $A = 20 \log \left[\frac{k}{10K} \right] = -20 \text{ dB}$



From the above analysis, it is evident that magnitude plot of the factor is a straight line with a slope of -20 dB/decade and passing through 0 dB when $\omega = K$.

Since the angle $G(j\omega)$ is a constant and independent of ω , the phase plot is a straight line at -90° .

$\rightarrow$ when an integral factor has multiplicity of n

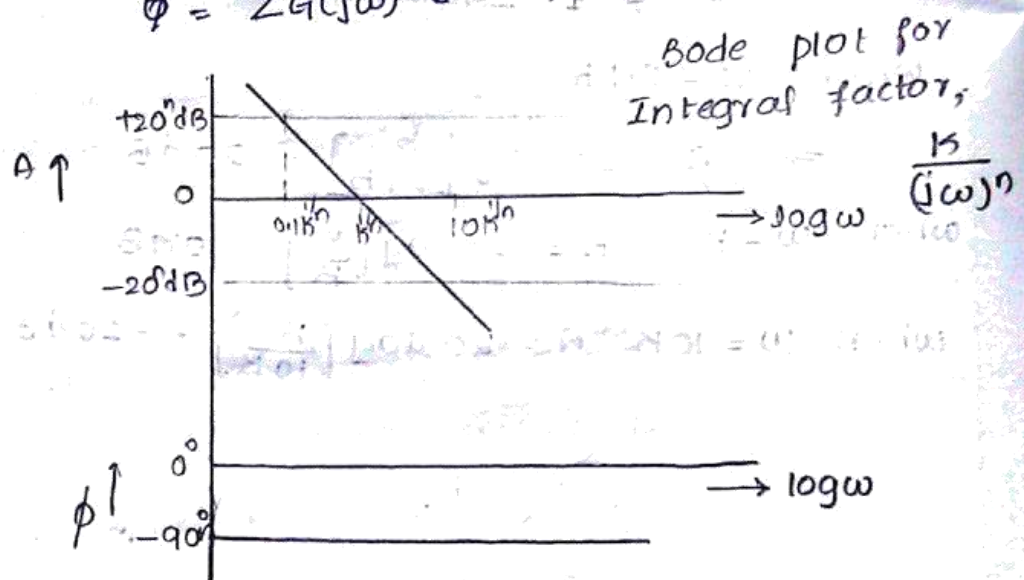
then $G(s) = \frac{k}{s^n}$

$\Rightarrow G(j\omega) = \frac{k}{(j\omega)^n}$

$A = |G(j\omega)| = \frac{k}{\omega^n} \angle -90^\circ$

$A = |G(j\omega)|_{dB} = 20 \log \left| \frac{k}{\omega^n} \right|$

$\phi = \angle G(j\omega) = -90^\circ$



Derivative Factor :

$G(s) = ks$

$G(j\omega) = k j\omega$

$|G(j\omega)| = k\omega$

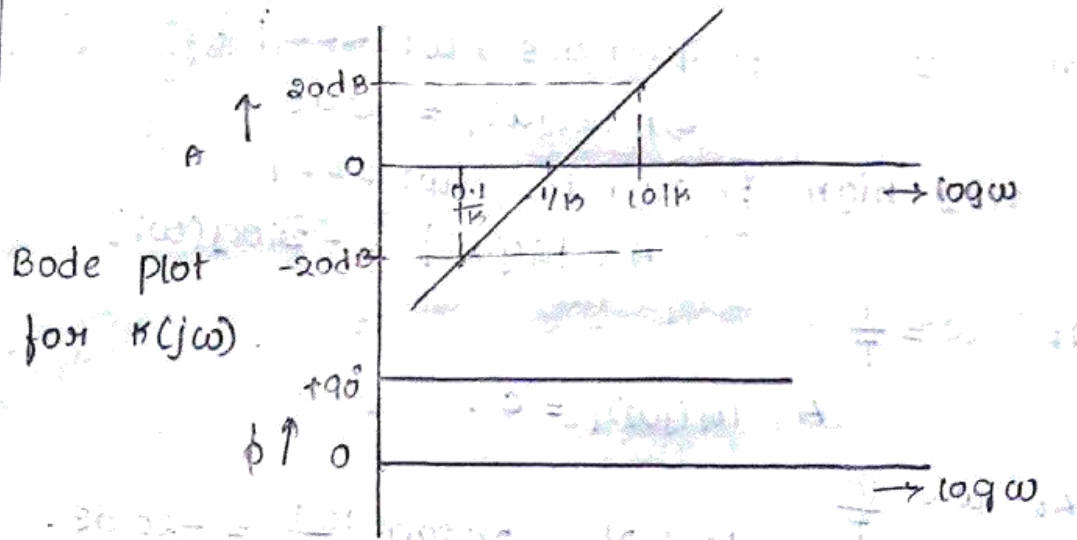
$\phi = \angle G(j\omega) = +90^\circ$

$A = |G(j\omega)|_{dB} = 20 \log(k\omega)$

when $\omega = \frac{0.1}{k} \Rightarrow A = 20 \log \left[\frac{0.1 k}{k} \right] = -20$

$\omega = \frac{1}{k} \Rightarrow A = 20 \log 1 = 0$

$\omega = \frac{10}{k} \Rightarrow A = 20 \log[10] = 20$



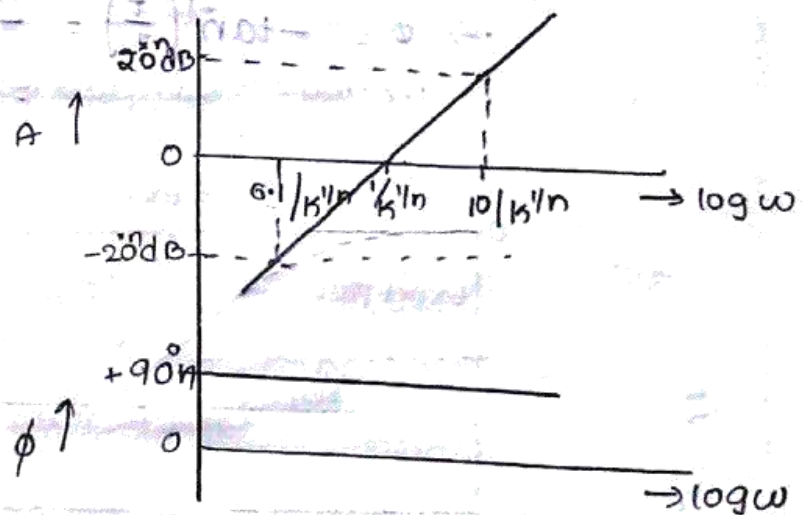
For $G(s) = Ks^n$

$$G(j\omega) = K(j\omega)^n$$

$$|G(j\omega)| = K\omega^n$$

$$\phi = \angle G(j\omega) = 90^\circ$$

$$A = |G(j\omega)|_{\text{in dB}} = 20 \log |K\omega^n|$$



First order factor in denominator :

$$G(s) = \frac{1}{1+sT}$$

$$G(j\omega) = \frac{1}{1+j\omega T}$$

$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2 T^2}} \cdot \tan^{-1}(\omega T)$$

$$\phi = \angle G(j\omega) = \frac{1}{\tan^{-1}(\omega T)} = -\tan^{-1}(\omega T)$$

$$A = |G(j\omega)| = 20 \log \left(\frac{1}{\sqrt{1+\omega^2 T^2}} \right) = -20 \log \sqrt{1+\omega^2 T^2}$$

At very low frequencies, $\omega T \ll 1$.

$$\Rightarrow A = |G(j\omega)| = 0 \text{ dB}$$

At very high frequencies, $\omega T \gg 1$

$$A = |G(j\omega)| = -20 \log(\omega T)$$

At $\omega = \frac{1}{T}$

$$A = |G(j\omega)| = 0$$

At $\omega = \frac{10}{T}$

$$A = |G(j\omega)| = -20 \log\left[\frac{10T}{T}\right] = -20 \text{ dB}$$

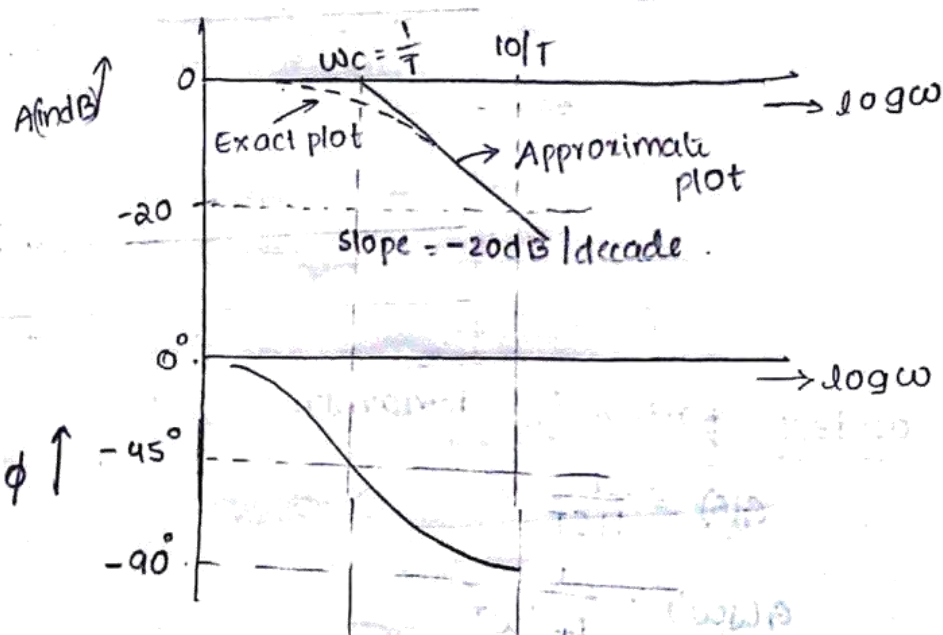
Phase $\phi = -\tan^{-1}(\omega T)$

when $\omega = 0$ $\phi = 0^\circ$

$\omega = \infty$ $\phi = -90^\circ$

At corner frequency $\omega_c = \frac{1}{T}$

$$\Rightarrow \phi = -\tan^{-1}\left(\frac{T}{T}\right) = -\tan^{-1}(1) = -45^\circ$$



Above analysis shows that magnitude plot of the factor $\frac{1}{1+j\omega T}$ can be approximated by two straight lines.

one is a straight line at 0 dB for frequency range $0 < \omega < \frac{1}{T}$ & other is a straight line

with slope -20dB/decade for frequency range

$$\frac{1}{T} < \omega < \infty$$

Two straight lines are Asymptotes of exact curve.

The frequency at which 2 Asymptotes meet is called corner frequency or break frequency.

For the factor $\frac{1}{1+j\omega T}$, $\omega = \frac{1}{T}$ is the corner frequency, ω_c .

It divides the curve into two regions - A curve for upper frequencies is a curve for lower frequencies.

Actual magnitude at corner frequency $\omega_c = \frac{1}{T}$ is

$$|G(j\omega)| = -20 \log \sqrt{1+\omega^2 T^2}$$
$$= -20 \log \sqrt{2} = -3.01 \text{ dB}$$

Hence, by this approximation, the loss in dB at corner frequency is -3dB

First order factor in numerator:

$$G(s) = 1 + sT$$

$$G(j\omega) = 1 + j\omega T$$

$$|G(j\omega)| = \sqrt{1+\omega^2 T^2}$$

$$\phi = \angle G(j\omega) = \tan^{-1}(\omega T)$$

$$A = |G(j\omega)| = 20 \log \sqrt{1+\omega^2 T^2}$$

At lower frequencies $\omega T \ll 1$

$$A = 0$$

At very high frequencies $\omega T \gg 1$

$$A = 20 \log(\omega T)$$

$$\text{at } \omega = \frac{1}{T} \quad A = 0 \text{ dB}, \quad \omega = \frac{10}{T} = 20 \text{ dB.}$$

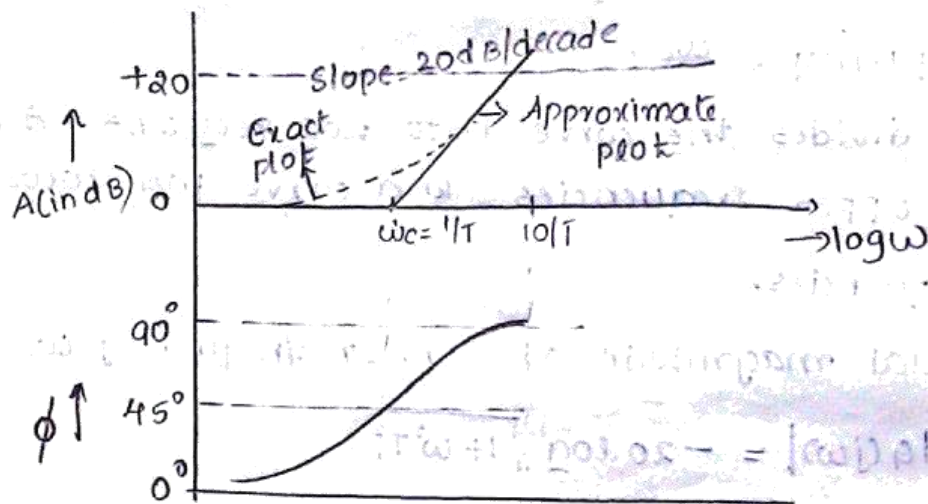
$$\text{Phase } \phi = \tan^{-1}[\omega T]$$

$$\text{when } \omega = 0, \phi = 0^\circ$$

$$\omega = \infty, \phi = 90^\circ$$

$$\text{At corner frequency } \omega_c = \frac{1}{T}$$

$$\phi = \tan^{-1}\left[\frac{T}{T}\right] = \tan^{-1}(1) = 45^\circ$$



Actual magnitude at corner frequency $\omega_c = \frac{1}{T}$

$$\text{is } |G(j\omega)| = 80 \log \sqrt{1 + \omega^2 T^2}$$

$$= 80 \log \sqrt{2}$$

$$= 3.01 \text{ dB.}$$

Quadratic factor in Denominator:

$$G(s) = \frac{s^2 \omega_n^2}{s^2 + 2\xi \omega_n s + \omega_n^2}$$

$$G(j\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\xi \omega_n j\omega + \omega_n^2}$$

$$G(j\omega) = \frac{1}{-\frac{\omega^2}{\omega_n^2} + 2j\xi \frac{\omega}{\omega_n} + 1}$$

$$\Rightarrow G(j\omega) = \frac{1}{\left(1 - \frac{\omega^2}{\omega_n^2}\right) + j\left(2\xi\frac{\omega}{\omega_n}\right)}$$

$$|G(j\omega)| = 20 \log \left[\frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}} \right]$$

$$A = -20 \log \left[\sqrt{\left(1 - \frac{\omega^2}{\omega_n^2}\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2} \right]$$

$$\angle G(j\omega) = -\tan^{-1} \left[\frac{2\xi\omega/\omega_n}{\left(1 - \frac{\omega^2}{\omega_n^2}\right)} \right]$$

* Problems :

1. Draw the bode plot for the given Transfer function

$$G(s) = \frac{80}{s(s+2)(s+20)}$$

Sol:-

$$G(s) = \frac{80}{s(s+2)(s+20)}$$

$$= \frac{80}{2(20)(s)\left(1+\frac{s}{2}\right)\left(1+\frac{s}{20}\right)}$$

$$= \frac{2}{s\left(1+\frac{s}{2}\right)\left(1+\frac{s}{20}\right)}$$

Replace s by ' $j\omega$ '

$$\Rightarrow G(j\omega) = \frac{2}{j\omega\left(1+j\frac{\omega}{2}\right)\left(1+j\frac{\omega}{20}\right)}$$

$$\text{Here } T_1 = \frac{1}{2} \quad T_2 = \frac{1}{20}$$

$$\text{corner frequencies } \omega_{c1} = \frac{1}{T_1} = 2 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{T_2} = 20 \text{ rad/sec}$$

Term.	Corner frequency (rad/sec)	slope (dB/decade)	change in slope (dB/decade)
$\frac{2}{j\omega}$		-20	-20
$\frac{1}{1+j(0.5)\omega}$	$\omega_{c1} = 2 \text{ rad/sec}$	-20	-40
$\frac{1}{1+j(0.05)\omega}$	$\omega_{c2} = 20 \text{ rad/sec}$	-20	-60

$$\text{Let } \omega_L = 0.2, \quad \omega_H = 200$$

$$\text{If } \omega = \omega_L, \quad A = 20 \log \left| \frac{2}{j\omega} \right| = 20 \log \left| \frac{2}{j0.2} \right|$$

$$= 20 \log \left| \frac{2}{0.2j} \right| = 20 \text{ dB}$$

At $\omega = \omega_{c1}$, $A = 20 \log \left| \frac{2}{2} \right| = 0$.

At $\omega = \omega_{c2}$, $A = 20 \log \left| \frac{2}{20} \right| = -20 \text{ dB}$

Gain = (slope from ω_{c1} to ω_{c2} $\times \log \left\{ \frac{\omega_{c2}}{\omega_{c1}} \right\}$)
 + (Gain A at $\omega = \omega_{c1}$)

= $-40 \times \log \left[\frac{20}{2} \right] + 0$

= $-40 \times \log [1] = -40 \text{ dB}$

At $\omega = \omega_h$,

Gain = (slope from ω_{c2} to $\omega_h \times \log \left[\frac{\omega_h}{\omega_{c2}} \right]$
 + Gain A at $\omega = \omega_{c2}$.

= $-60 \times \log \left[\frac{200}{20} \right] + (-40)$

= $-60 \times (\log 10) - 40 = -100 \text{ dB}$.

Phase

$\phi = \angle G(j\omega) = \frac{\angle (2+j0)}{\angle j\omega \cdot \angle 1+j\frac{\omega}{2} - \angle 1+j\frac{\omega}{20}}$

= $\frac{0^\circ}{90^\circ + \angle \tan^{-1} \left(\frac{\omega}{2} \right) - \angle \tan^{-1} \left(\frac{\omega}{20} \right)}$

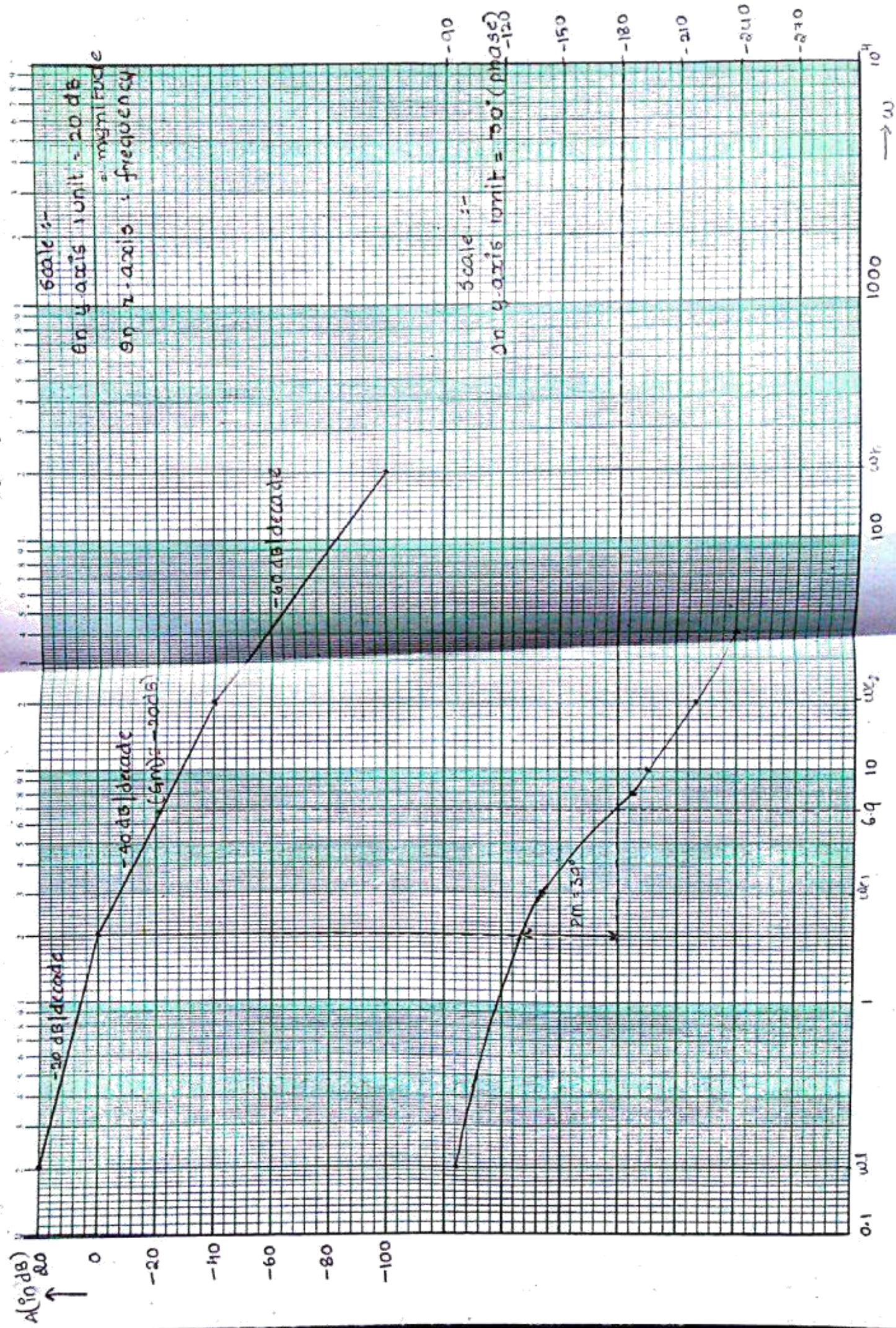
$\therefore \phi = -90^\circ - \tan^{-1} \left(\frac{\omega}{2} \right) - \tan^{-1} \left(\frac{\omega}{20} \right)$

ω	$\angle j\omega$	$-\tan^{-1} \left(\frac{\omega}{2} \right)$	$-\tan^{-1} \left(\frac{\omega}{20} \right)$	ϕ
0.2	-90°	-5.71	-0.57	-96.28°
2	-90°	-45°	-5.71	-140.71°
8	-90°	-75.96	-21.80	-187.76°
10	-90°	-78.69	-26.56	-195.25°
20	-90°	-84.28	-45°	-219.28°
40	-90°	-87.13	-63.43	-240.56°
∞	-90°	-90°	-90°	-270°

$$G(s) = \frac{80}{s(s+2)(s+20)}$$

1) Bode plot for Transfer function

Key 1 SEMI-LOG PAPER (5 CYCLES x 1/10°)



$$\omega_{gc} \text{ (gain cross over frequency)} = 2 \text{ rad/sec.}$$

$$\text{Phase margin (PM)} = 180 - 141 = 39^\circ.$$

$$\omega_{pc} \text{ (phase cross over frequency)} = 6.3 \text{ rad/sec.}$$

$$\text{Gain margin (GM)} \approx 20 \text{ dB}$$

✓ system is stable

2. Sketch the bode plot for the following transfer

$$\text{function } G(s) = \frac{s^2}{(1+0.2s)(1+0.02s)}$$

$$\text{soln: } G(s) = \frac{s^2}{(1+0.2s)(1+0.02s)}$$

$$G(j\omega) = \frac{(j\omega)^2}{(1+0.2j\omega)(1+0.02j\omega)}$$

$$T_1 = 0.2\omega, \quad T_2 = 0.02\omega.$$

$$\text{Corner frequencies, } \omega_{c1} = \frac{1}{T_1} = \frac{1}{0.2} = 5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{T_2} = \frac{1}{0.02} = 50 \text{ rad/sec}$$

Term	corner frequencies	slope (dB/decade)	change in slope (in dB/decade)
$(j\omega)^2$	90°	$20 \times 2 = 40 \text{ dB/dec}$	40
$\frac{1}{1+0.2j\omega}$	$\omega_{c1} = 5 \text{ rad/sec}$	-20	20
$\frac{1}{1+0.02j\omega}$	$\omega_{c2} = 50 \text{ rad/sec}$	-20	0

$$\text{Let } \omega_l = 0.5, \quad \omega_h = 500$$

$$\text{At } \omega = \omega_l, \quad A = 20 \log |(j\omega)^2| = 40 \log (0.5) = -12.041$$

$$\text{At } \omega = \omega_{c1}, \quad A = 20 \log (j(5)^2) = 27.95$$

$$\text{At } \omega = \omega_{c2}$$

$$\text{Gain} = (\text{slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \left(\frac{\omega_{c2}}{\omega_{c1}} \right) + \text{Gain at } \omega = \omega_{c1})$$

$$\omega_{gc} \text{ (gain cross over frequency)} = 2 \text{ rad/sec.}$$

$$\text{Phase margin (PM)} = 180 - 141 = 39^\circ$$

$$\omega_{pc} \text{ (phase cross over frequency)} = 6.3 \text{ rad/sec.}$$

$$\text{Gain margin (GM)} \approx 20 \text{ dB}$$

✓ system is stable

2. Sketch the bode plot for the following transfer

$$\text{function } G(s) = \frac{s^2}{(1+0.2s)(1+0.02s)}$$

sol:-

$$G(s) = \frac{s^2}{(1+0.2s)(1+0.02s)}$$

$$G(j\omega) = \frac{(j\omega)^2}{(1+0.2j\omega)(1+0.02j\omega)}$$

$$T_1 = 0.2\omega, \quad T_2 = 0.02\omega$$

$$\text{Corner frequencies, } \omega_{c1} = \frac{1}{T_1} = \frac{1}{0.2} = 5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{T_2} = \frac{1}{0.02} = 50 \text{ rad/sec}$$

Term	corner frequencies	slope (dB/decade)	change in slope (in dB/decade)
$(j\omega)^2$	90°	$20 \times 2 = 40 \text{ dB/dec}$	40
$\frac{1}{1+0.2j\omega}$	$\omega_{c1} = 5 \text{ rad/sec}$	-20	20
$\frac{1}{1+0.02j\omega}$	$\omega_{c2} = 50 \text{ rad/sec}$	-20	0

$$\text{Let } \omega_l = 0.5, \quad \omega_h = 500$$

$$\text{At } \omega = \omega_l, \quad A = 20 \log |(j\omega)^2| = 40 \log (0.5) = -12.041$$

$$\text{At } \omega = \omega_{c1}, \quad A = 20 \log (j(5)^2) = 27.95$$

$$\text{At } \omega = \omega_{c2}$$

$$\text{Gain} = (\text{slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \left(\frac{\omega_{c2}}{\omega_{c1}} \right) + \text{Gain at } \omega = \omega_{c1})$$

$$20 \times \log \left| \frac{50}{5} \right| + 27.95 = 47.9$$

At $\omega = \omega_h$

slope from ω_{c2} to $\omega_h \times \log \left| \frac{\omega_h}{\omega_{c2}} \right| +$
 gain at $\omega = \omega_{c2}$

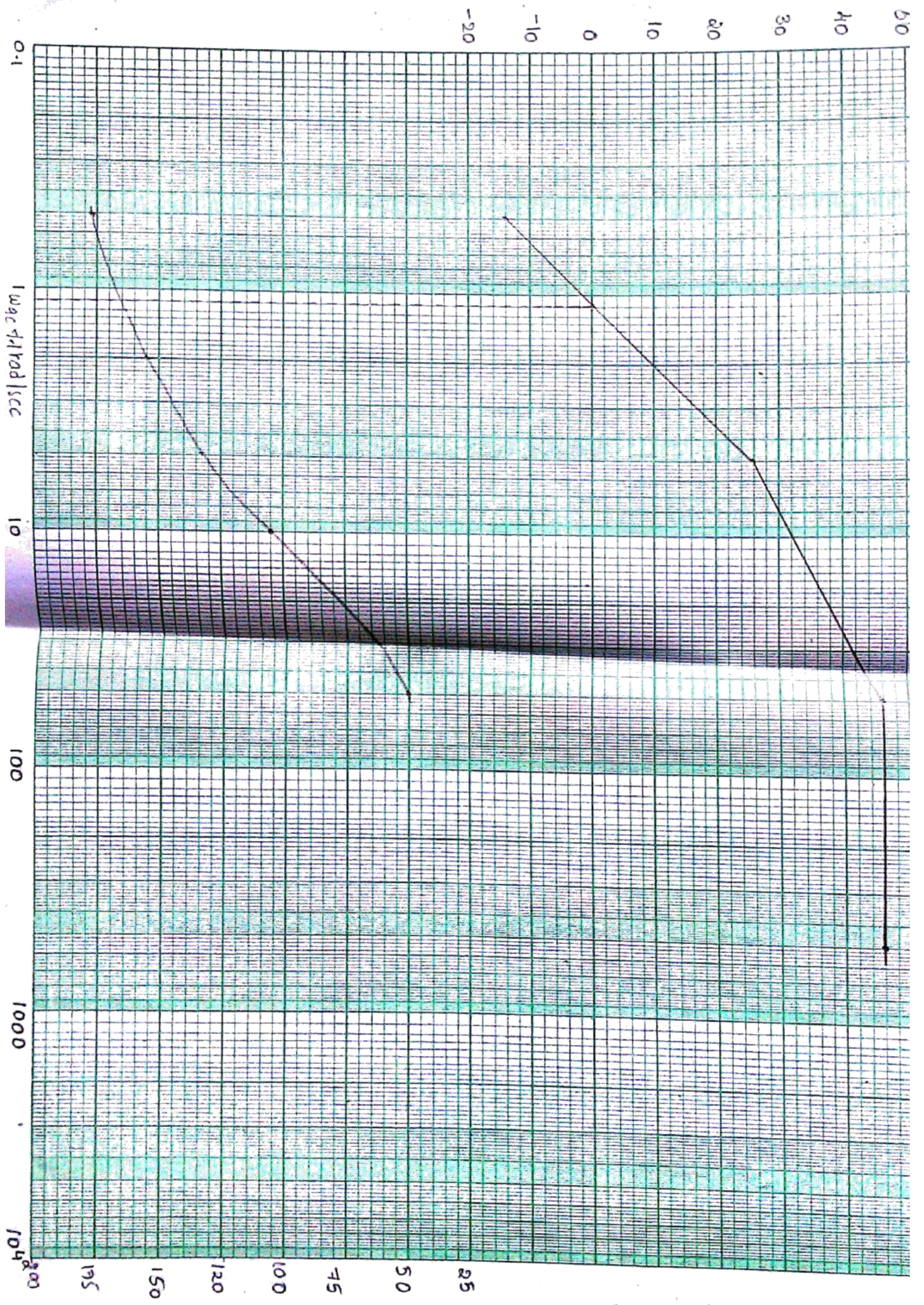
$$= 0 \times \log \left| \frac{500}{50} \right| + 47.9 = 47.9$$

$$\text{Phase } \angle G(j\omega) = \frac{\angle (j\omega)^2}{\angle (1+0.2j\omega) \cdot \angle (1+0.02j\omega)}$$

$$= \frac{180^\circ}{\tan^{-1}(0.2\omega) \cdot \tan^{-1}(0.02\omega)}$$

$$\phi = 180^\circ - \tan^{-1}(0.2\omega) - \tan^{-1}(0.02\omega)$$

ω	$(j\omega)^2$	$-\tan^{-1}(0.2\omega)$	$-\tan^{-1}(0.02\omega)$	ϕ
0.5	180°	-5.71	-0.572	173.72
2	180°	-21.80	-2.29	155.91
5	180°	-45°	-5.71	129.29
10	180°	-63.43	-11.30	105.27
50	180°	-84.28	-45	50.72
∞	180°	-90°	-90°	0



* Calculation of Gain margin and Phase margin from Bode plot :

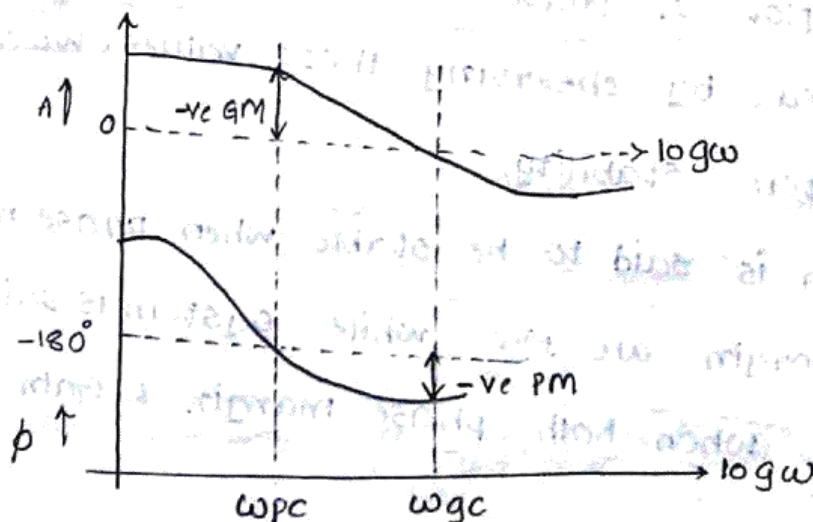
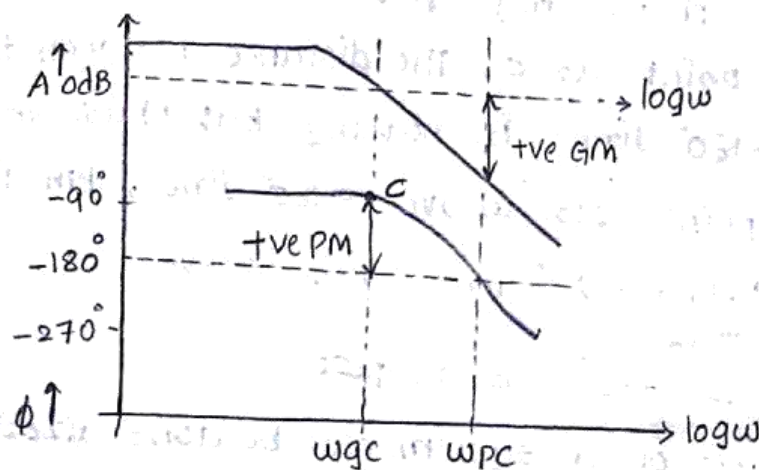
According to definition of gain margin

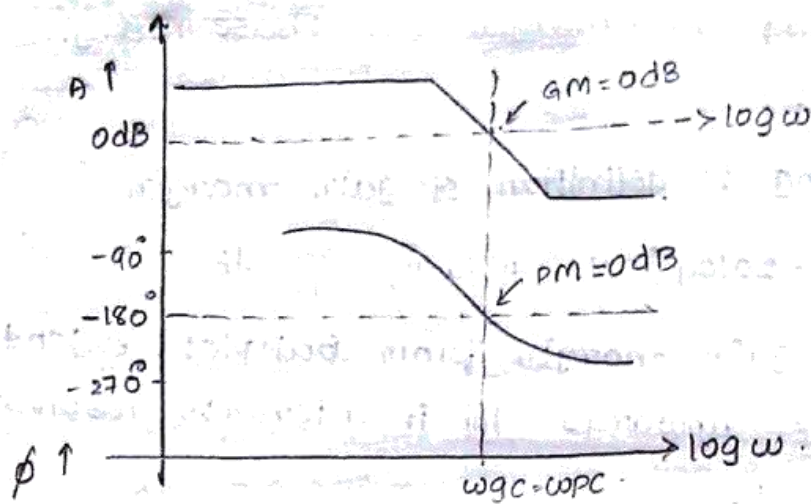
$$GM = -20 \log |G(j\omega) H(j\omega)|_{\omega=\omega_{pc}} \text{ dB}$$

To find gain margin from Bodeplot, extend $\omega = \omega_{pc}$ line upwards till it intersects resultant magnitude plot.

The magnitude corresponding to this intersection point is gain margin.

If intersection point is below 0dB, gain margin is possible and if point is above 0dB, gain margin is negative





Phase margin :

$$\text{Phase margin} = [\angle G(j\omega)]_{\omega=\omega_{gc}} - (-180^\circ)$$

If phase $[\angle G(j\omega)]_{\omega=\omega_{gc}}$ can be read out from phase plot. Now extend $\omega=\omega_{gc}$ line downwards till it intersects phase angle plot.

Assume this point as c. The distance between this point c and -180° line is nothing but phase margin. If intersection point c is above -180° line, PM is +ve and if c is below -180° line, PM is -ve.

* Stability Analysis from Bode plot :

Stability Analysis of a system can be done directly from Bode plot. If phase margin & Gain margin are calculated, by observing these values, we can analyse system stability.

System is said to be stable when phase margin & Gain margin are +ve while system is said to be unstable when both phase margin & Gain margin are -ve.

system is said to be marginally stable when ζ or γ are zero.

- Quadratic factor in Denominator

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

slope = -40 dB/decade

- Quadratic factor in Numerator

$$G(s) = \frac{s^2 + 2\zeta\omega_n s + \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

slope = +40 dB/decade.

- Q. Sketch the bode plot for following transfer function
 & Determine phase margin & gain margin.

$$G(s) = \frac{75(1+0.2s)}{s(s^2+16s+100)}$$

Sol: Comparing deno. term $s^2+16s+100$ with $s^2+2\zeta\omega_n s+\omega_n^2$
 $\Rightarrow \omega_n^2 = 100 \Rightarrow \omega_n = 10.$

$$G(s) = \frac{75(1+0.2s)}{s(100) \left[1 + \frac{16s}{100} + \frac{s^2}{100} \right]}$$

replace s by $j\omega$

$$\Rightarrow G(j\omega) = \frac{75(1+0.2j\omega)}{j\omega(100) [1 + 0.16j\omega - 0.01\omega^2]}$$

$$G(j\omega) = \frac{0.75(1+0.2j\omega)}{j\omega(1 - 0.01\omega^2 + j0.16\omega)}$$

$$|G(j\omega)| = 0.75$$

At $\tau = 0.2$ sec, $\omega_{c1} = \frac{1}{0.2} = 5 \text{ rad/sec}$

$$\omega_n = \omega_{c2} = 10 \text{ rad/sec}$$

Term	corner frequency	slope	change in slope
$\frac{0.75}{j\omega}$		-20	-20
$1+0.2j\omega$	$\omega_{c1}=5 \text{ rad/sec}$	20	-40
$\frac{1}{1-0.01\omega^2 + j0.16\omega}$	$\omega_{c2}=10 \text{ rad/sec}$	-40	-40

Let $\omega_l = 0.5 \text{ rad/sec}$, $\omega_h = 100 \text{ rad/sec}$.

$$\text{At } \omega = \omega_l = 0.5 \quad A = 20 \log \left| \frac{0.75}{j\omega} \right|$$

$$= 20 \log \left| \frac{0.75}{0.5} \right| = 3.52.$$

$$\text{At } \omega = \omega_{c1} = 5 \text{ rad/sec} \quad A = 20 \log \left| \frac{0.75}{5} \right| = -16.47$$

$$\text{At } \omega = \omega_{c2}$$

$$A = \text{slope from } \omega_{c1} \text{ to } \omega_{c2} \times \log \left| \frac{\omega_{c2}}{\omega_{c1}} \right| +$$

$$\text{Gain } A \text{ at } \omega = \omega_{c1}$$

$$= 20 \times \log \left| \frac{10}{5} \right| - 16.47 =$$

$$= -16.47$$

$$\text{At } \omega = \omega_h = 100$$

$$A = \text{slope from } \omega_{c2} \text{ to } \omega_h \times \log \left| \frac{\omega_h}{\omega_{c2}} \right| +$$

$$\text{Gain at } \omega = \omega_{c2}$$

$$= -40 \times \log \left| \frac{100}{10} \right| + (-16.47)$$

$$= -56.47$$

$$\text{Phase } \angle G(j\omega) = \frac{\angle(1+j0.2\omega)}{j\omega \angle(1-0.01\omega^2 + j0.16\omega)}$$

$$\Rightarrow \angle G(j\omega) = \frac{\tan^{-1}(0.2\omega)}{\angle 90^\circ + \angle \tan^{-1} \left(\frac{0.16\omega}{1-0.01\omega^2} \right)}$$

$$\Rightarrow \angle G(j\omega) = \tan^{-1}(0.2) - 90^\circ - \tan^{-1}\left(\frac{0.16\omega}{1-0.01\omega^2}\right)$$

ω	$1/j\omega$	$\tan^{-1}(0.2)$	$-\tan^{-1}\left(\frac{0.16\omega}{1-0.01\omega^2}\right)$	ϕ
0.5	-90°	5.71	-4.58	-100.29 - 88.87
2	-90°	21.80	-18.43	-86.63
5	-90°	45	-40.84	-91.84
10	-90°	63.43	-90°	-116.57
100	-90°	87.13	+9.18 - 180	8.21 -173.69
20	-90°	75.96	-89.67 -133.15	-147.19
50	-90°	84.28	-161.56	-167.28
∞	-90°	90°	-270°	-270.

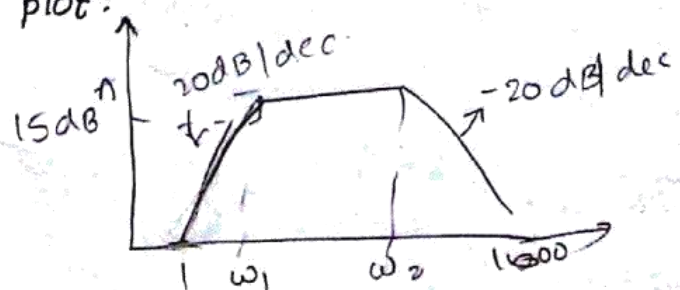
- * Calculation of transfer function from magnitude plot:
- Starting slope of magnitude plot represents poles of zeros at the origin.
- If starting slope is -20 dB/decade , there is one pole at origin.
- If starting slope is -40 dB/decade , there are 2 poles at origin.
- If starting slope is 0 dB/decade , there is no pole or no zero at origin.
- If starting slope is 20 dB/decade , there is one zero at origin.
- Then observe the shift in magnitude plot at $\omega=1$, which represents $20 \log k$ from which we can decide value of k . Then this slope changes at each corner frequency. So identify corner frequencies. The change in slope indicates respective factor with corresponding corner frequencies.

If change in slope is -20 dB/decade , then the factor is simple pole.

If change in slope is $+20 \text{ dB/decade}$, then the factor is simple zero.

Taking the product of all of them we can construct the Transfer Function of the given system.

1. Determine the transfer function of the system from the following plot.



Sol:-

ω	^a Initial slope	^b 2 nd slope	change in slope = (b-a)	Factor
$\omega_{c1} = 1$	0 dB/dec	20 dB/dec	(20-0) = 20 dB	$1+ST_1$
$\omega_{c2} = \omega_1$	20 dB/dec	0 dB/dec	-20 dB/dec	$\frac{1}{1+ST_2}$
$\omega_{c3} = \omega_2$	0	-20	-20 dB/dec	$\frac{1}{1+ST_3}$
$\omega_{c4} = 1000$	-20	0	20 dB/dec	$1+ST_4$

$$1+ST_1 :$$

$$T_1 = \frac{1}{\omega_{c1}} \Rightarrow T_1 = 1 \quad \text{Factor} = (1+s)$$

$$\frac{1}{1+ST_2} :$$

$$T_2 = \frac{1}{\omega_{c2}} = \frac{1}{\omega_1}$$

To find ω_1 , write sk. line eqn $y = mx + c_1$

$$y = 15 \text{ dB}, \quad m = 20 \text{ dB/dec}, \quad x = \log \omega_1$$

$$15 = 20 \log \omega_1 + c_1 \rightarrow \textcircled{1}$$

$$y = 0 \text{ dB}, \quad m = 20 \text{ dB/dec}, \quad x = \log 1$$

$$0 = 20 \log 1 + c_1 \Rightarrow c_1 = 0 \rightarrow \textcircled{2}$$

sub $\textcircled{2}$ in $\textcircled{1}$

$$20 \log \omega_1 = 15$$

$$\omega_1 = 15.623 \Rightarrow T_2 = \frac{1}{\omega_1} = 0.177$$

$$\frac{1}{1+0.177s} \quad (\text{factor 2})$$

$$\frac{1}{1+ST_3} :$$

$$y = mx + c_2 \quad T_3 = \frac{1}{\omega_{c3}} = \frac{1}{\omega_2}$$

$$y = 15, \quad m = -20, \quad x = \log \omega_2$$

$$15 = -20 \log \omega_2 + c_2 \rightarrow \textcircled{1}$$

$$0 = -20 \log 1000 + c_2$$

$$c_2 = 20 \log 10^3$$

$$c_2 = 60 \text{ dB} \rightarrow \textcircled{2}$$

Subs. ② in ①

$$15 = -20 \log \omega_2 + 60$$

$$\frac{15}{20} = -\log \omega_2 + 3$$

$$T_3 = 0.000562, \quad \omega_2 = 177.83.$$

$1 + sT_4$:

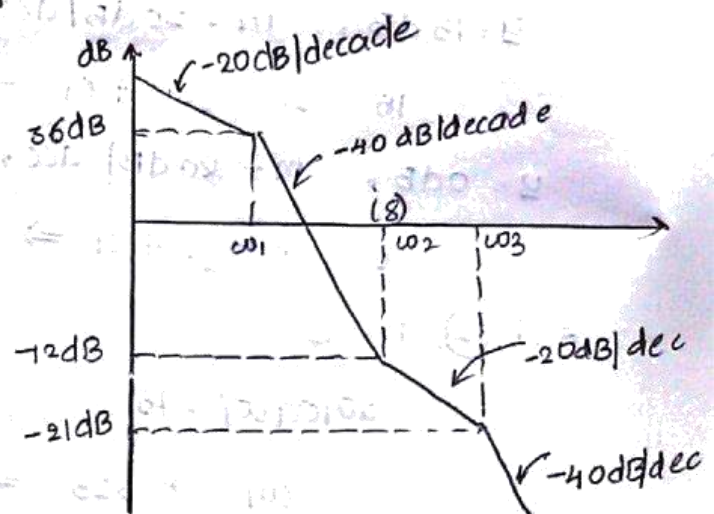
$$T_4 = \frac{1}{\omega_{c4}} = \frac{1}{1000} = 0.001$$

$$TF = \frac{(1+s)(1+0.001s)}{(1+s0.000562)(1+0.1777s)}$$

Gain : $20 \log K = 0$

$$K = 1$$

2.



ω	a Initial slope	b second slope	change in slope	Factor
ω_1	-20 dB/dec	-40 dB/dec	-20 dB/dec	$\frac{1}{1+sT_1}$
$\omega_2 = 8$	-40 dB/dec	-20 dB/dec	20 dB/dec	$1+sT_2$
ω_3	-20 dB/dec	-40 dB/dec	-20 dB/dec	$\frac{1}{1+sT_3}$

$$\frac{1}{1+sT_1} :$$

$$y = mx + c$$

$$y = 36 \text{ dB}, m = -40 \text{ dB/dec}, x = \log \omega_1$$

$$\Rightarrow 36 = -40 \log \omega_1 + c_1 \rightarrow \textcircled{1}$$

$$\text{ii)} y = -12, m = -40 \text{ dB}, x = \log \omega_2$$

$$\Rightarrow -12 = -40 \log 8 + c_1$$

$$c_1 = 24 \cdot 12$$

sub in $\textcircled{1}$

$$\Rightarrow 36 = -40 \log \omega_1 + c_1$$

$$\Rightarrow \frac{36 - 24 \cdot 12}{-40} = \log \omega_1$$

$$\omega_1 = 0.504$$

$$T_1 = \frac{1}{\omega_1} = \frac{1}{0.504} = 1.98$$

$$\Rightarrow \frac{1}{1+sT_1} = \frac{1}{1+(s)(1.98)}$$

$$1+sT_2 :$$

$$T_2 = \frac{1}{\omega_{c2}} = \frac{1}{8} = 0.125$$

$$\Rightarrow 1+s(0.125) = 1+sT_2$$

$$\frac{1}{1+sT_3} :$$

$$y = mx + c_2$$

$$y = -21 \text{ dB}, m = -20 \text{ dB}, x = \log \omega_3$$

$$-21 = -20 \log \omega_3 + c_2 \rightarrow \textcircled{3}$$

$$-21 = -20 \log \omega_3 + 24 \cdot 12$$

$$\omega_3 = 13.42$$

$$\text{at } y = -12$$

$$y = mx + C$$

$$-12 = -20 \log \omega_2 + C_2$$

$$-12 = -20 \log 8 + C_2$$

$$C_2 = -12 + 20 \log 8$$

$$C_2 = 6.06$$

sub in ③

$$-21 = -20 \log \omega_3 + C_2$$

$$-20 \log \omega_3 = -21 - 6.06$$

$$\omega_3 = 4.44 \cdot 22.54$$

$$T_3 = 0.044 \text{ s}$$

$$\Rightarrow \frac{1}{1+sT_3} = \frac{1}{1+0.044s}$$

$$20 \log \frac{K}{\omega_1} = 36$$

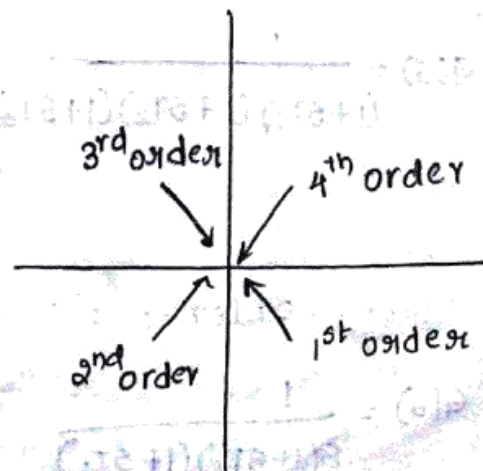
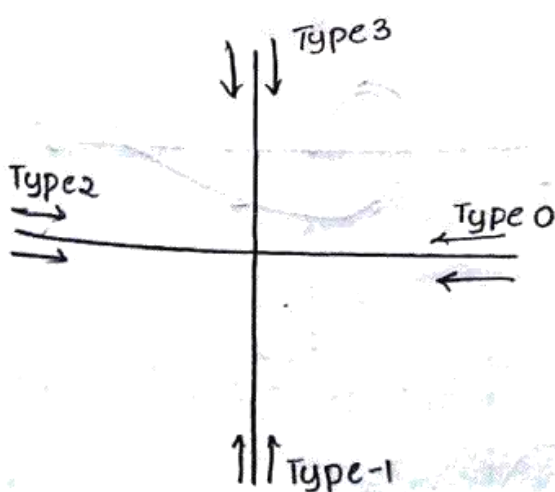
$$\log \frac{K}{\omega_1} = \frac{36}{20}$$

$$\Rightarrow K = 31.8$$

$$\Rightarrow T.F. = \frac{31.8 (1 + 0.125s)}{s (1 + 1.98s) (1 + 0.044s)}$$

POLAR PLOTS :

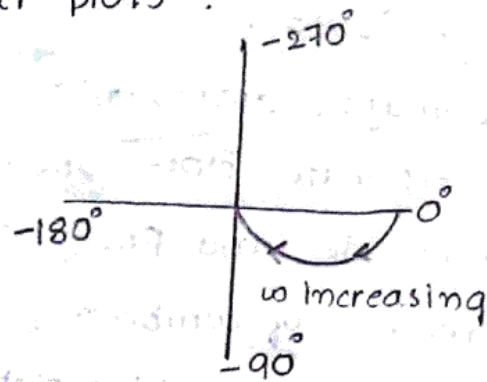
- The major disadvantage of Bode plot is that we have two separate plots showing the variation of magnitude and phase shift with frequency. A method of combining these 2 plots in a single plot is referred to as polar plot.
- Polar plot is a locus of vectors $|G(j\omega)| \angle G(j\omega)$ as ω is varied from 0 to ∞ .
- Polar plot is usually plotted on a polar graph sheet.
- Polar plot can be plotted on ordinary graph sheet. For this $G(j\omega)$ is expressed in rectangular form i.e., $G(j\omega) = G_R(j\omega) + jG_I(j\omega)$. Then polar plot is plotted between $G_R(j\omega)$ and $G_I(j\omega)$ as ω is varied from 0 to ∞ .
- For minimum phase transfer function with only poles, the type number of the system determine, at what quadrant, the polar plot starts and the order of the system determines, at what quadrant the polar plot ends.



* Typical sketches of polar plots :

i) Type-0, order-1 :

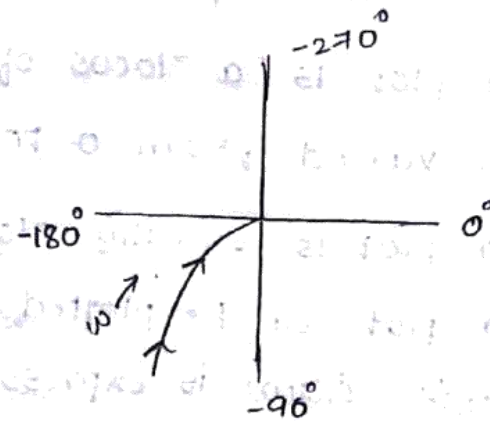
$$G(s) = \frac{1}{1+sT}$$



ii) Type-1, order-2 :

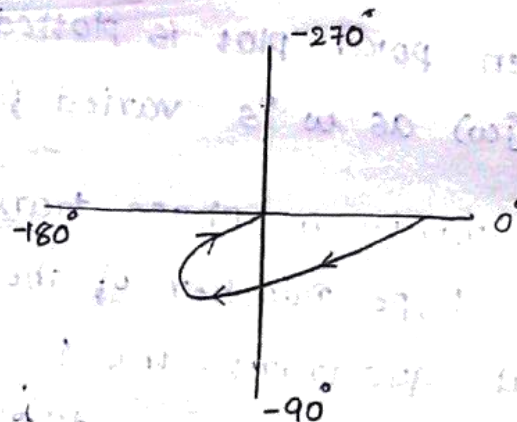
$$G(s) = \frac{1}{s(1+sT)}$$

$$G(j\omega) = \frac{1}{j\omega(1+j\omega T)}$$



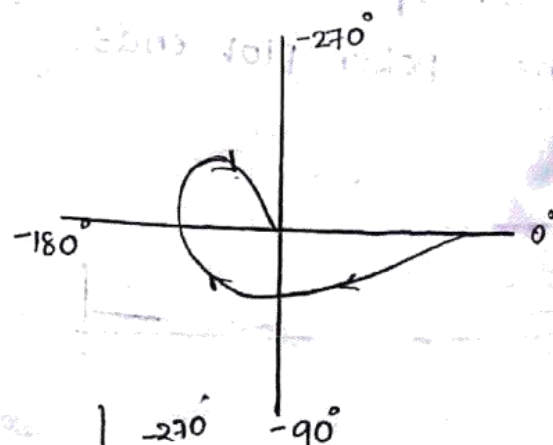
iii) Type-0, order-2 :

$$G(s) = \frac{1}{(1+sT_1)(1+sT_2)}$$



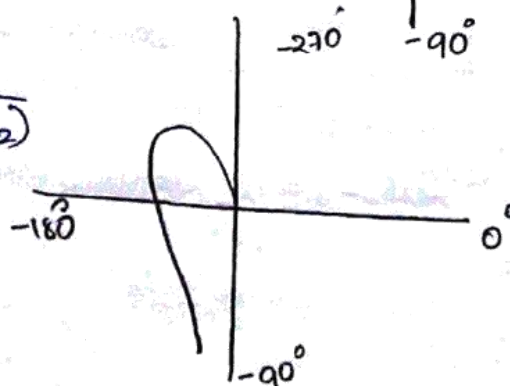
iv) Type-0, order-3 :

$$G(s) = \frac{1}{(1+sT_1)(1+sT_2)(1+sT_3)}$$



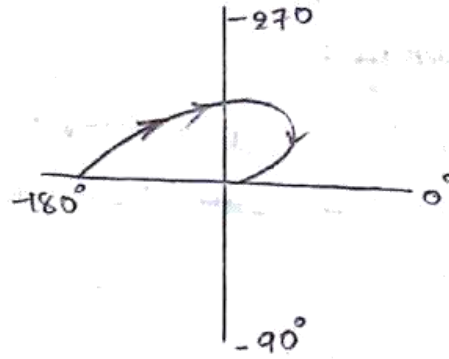
v) Type-1, order-3 :

$$G(s) = \frac{1}{s(1+sT_1)(1+sT_2)}$$



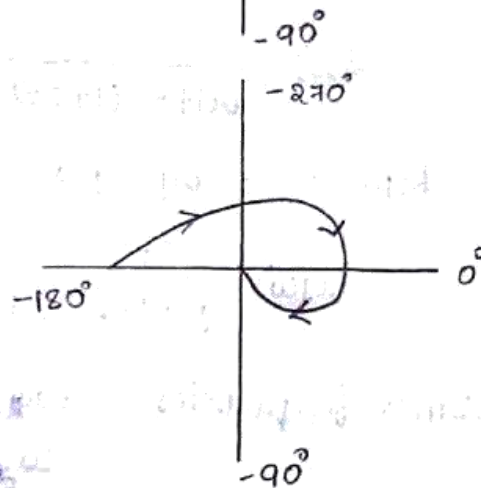
vi) Type-2, order-4 :

$$G(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)}$$



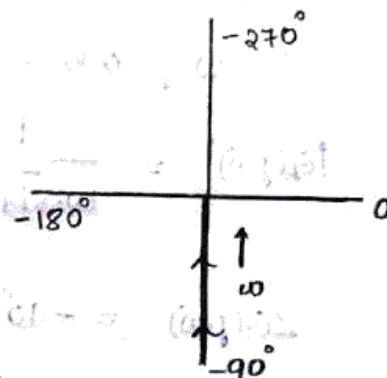
vii) Type-2, order-5 :

$$G(s) = \frac{1}{s^2(1+sT_1)(1+sT_2)(1+sT_3)}$$

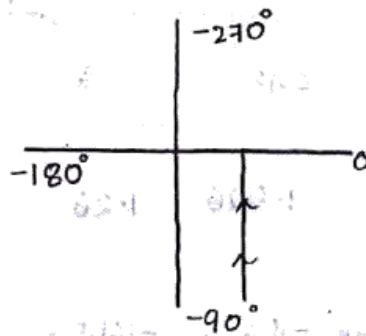


viii) $G(s) = \frac{1}{s}$ (Type-1)

$$G(j\omega) = \frac{1}{j\omega}$$

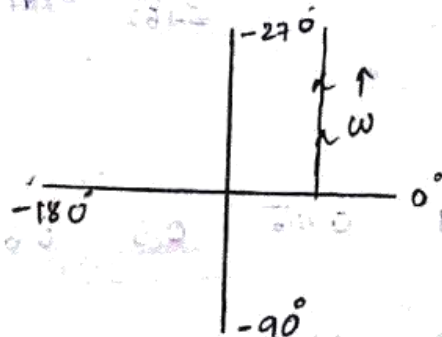


ix) $G(s) = \frac{1+sT}{sT}$



x) $G(s) = 1+sT$

$$G(j\omega) = 1+j\omega T$$



Problem :

1. The open loop transfer function of a unity feedback system is given by $G(s) = \frac{1}{s(1+s)(1+2s)}$. sketch the polar plot & determine gain margin and phase margin.

Sol:- $G(s) = \frac{1}{s(1+s)(1+2s)}$

Replace s by $j\omega$

$$G(j\omega) = \frac{1}{j\omega(1+j\omega)(1+2j\omega)}$$

corner frequencies $\omega_{c1} = 1$

$$\omega_{c2} = \frac{1}{2} = 0.5$$

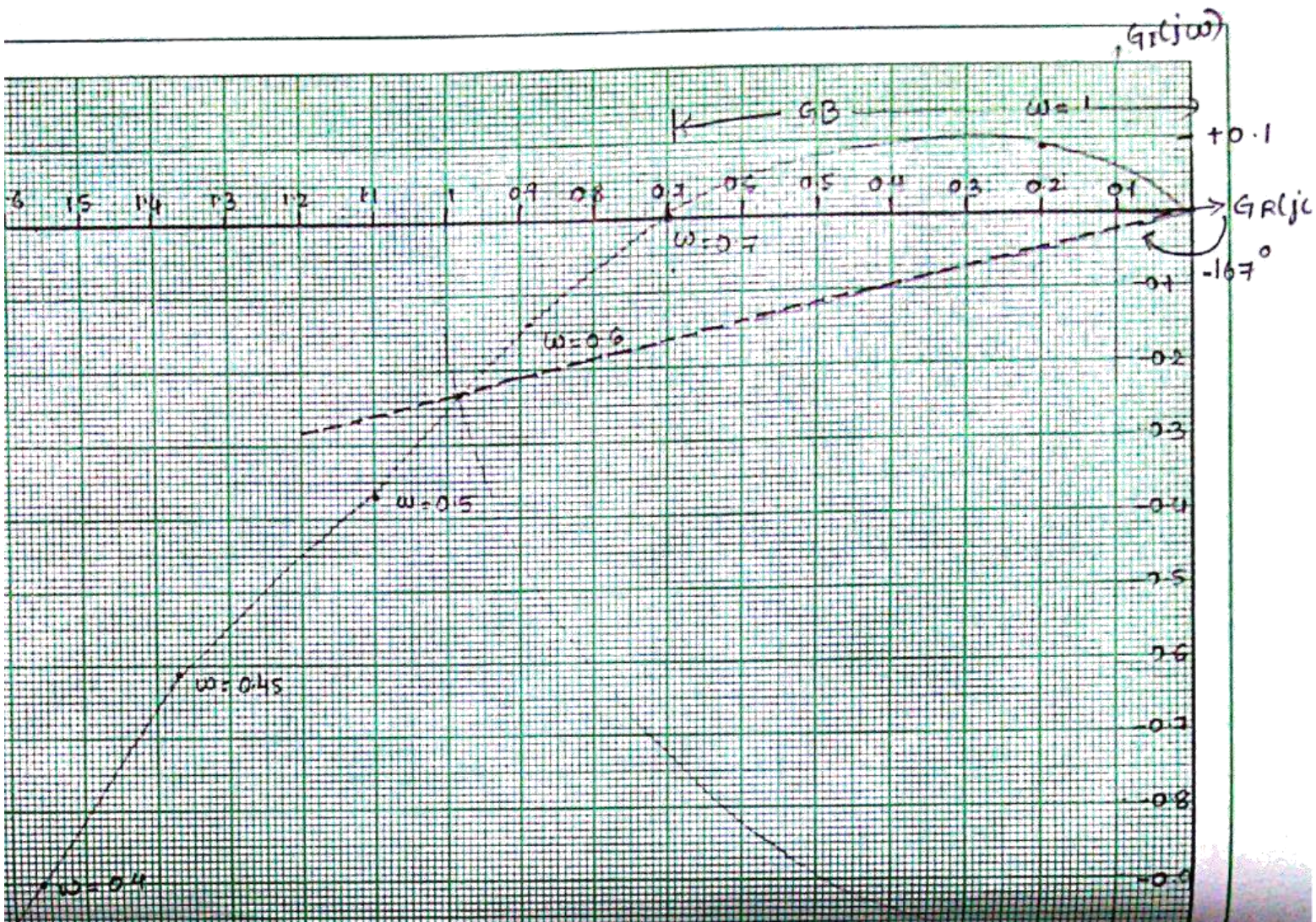
$$\Rightarrow \omega_{c1} = 0.5, \omega_{c2} = 1$$

$$|G(j\omega)| = \frac{1}{\omega \cdot \sqrt{1+\omega^2} \cdot \sqrt{1+(2\omega)^2}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(2\omega)$$

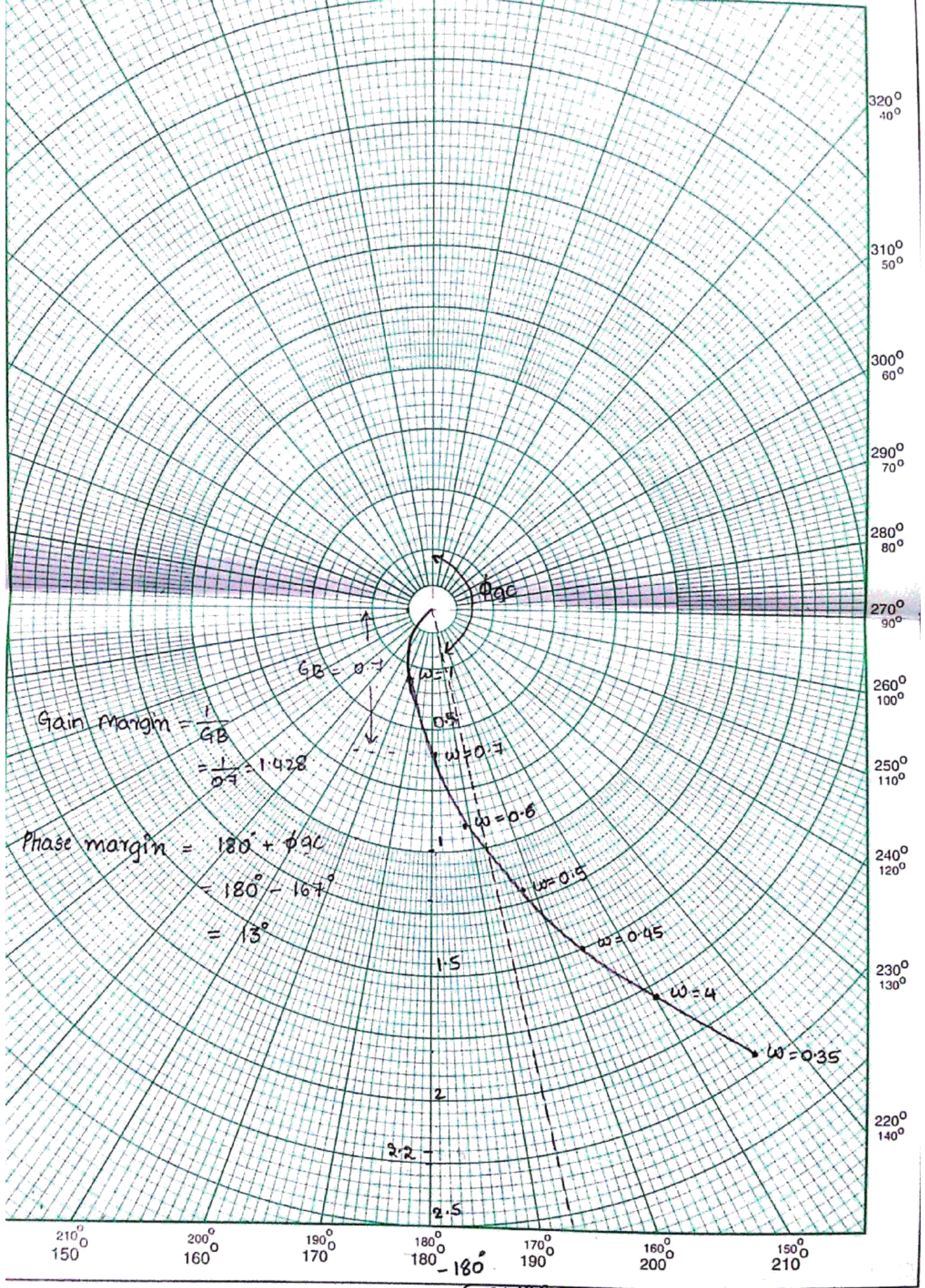
ω rad/sec	0.35	0.4	0.45	0.5	0.6	0.7	1
$ G(j\omega) $	2.209	1.81	1.506	1.26	0.91	0.68	0.31
$\angle G(j\omega)$	-144.28	-150.46	-156.21	-161.56 ≈ -162	-171.15	-179.45	-198.43

ω	0.35	0.4	0.45	0.5	0.6	0.7	1
$G_R(j\omega)$	-1.79	-1.57 1.56	-1.37	-1.19 1.19	-0.899	-0.679 0.7	-0.29
$G_I(j\omega)$	-1.28	-0.89 ≈ 0.9	-0.60 0.61	-0.39 0.39	-0.14	-7.12 × 10 ⁻³ ≈ 0	0.09



$$\text{Gain margin} = \frac{1}{GB} = \frac{1}{0.7} = 1.428$$

$$\begin{aligned} \text{Phase margin} &= 180 + \phi_{gc} \\ &= 180 - 16.7^\circ \\ &= 163.3^\circ \end{aligned}$$

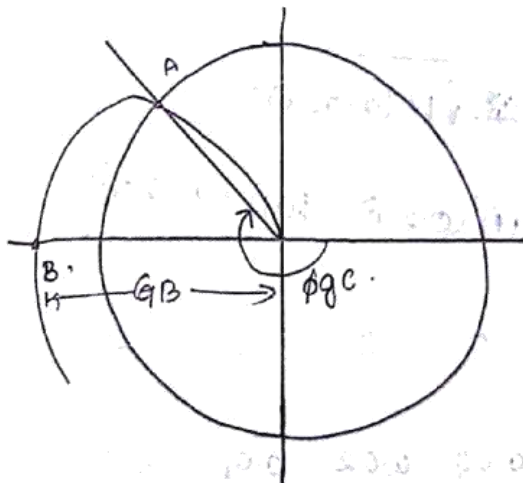


$$\text{Gain margin} = \frac{1}{G_B}$$

$$= \frac{1}{0.4} = 1.428$$

$$\begin{aligned} \text{Phase margin} &= 180^\circ + \phi_{gc} \\ &= 180^\circ - 167^\circ \\ &= 13^\circ \end{aligned}$$

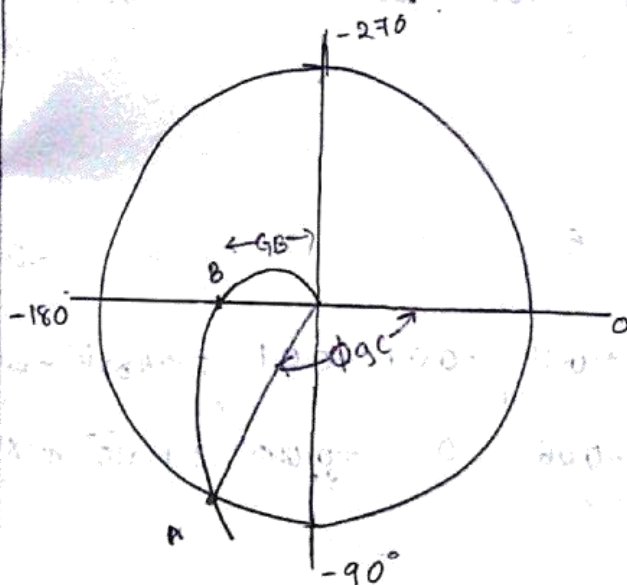
For Rectangular



$$GM = \frac{1}{G_B}$$

$$PM = 180^\circ + \phi_{gc}$$

For polar



$$GM = \frac{1}{G_B}$$

$$PM = 180^\circ + \phi_{gc}$$

2. consider a unity feedback system having an open loop transfer function $G(s) = \frac{k}{s(1+0.25s)(1+0.05s)}$. sketch the polar plot & determine the value of k so that gain margin is 18dB & phase margin is 60° .

Sol:

$$G(s) = \frac{k}{s(1+0.2s)(1+0.05s)}$$

replace 's' by 'jw'.

$$\Rightarrow G(j\omega) = \frac{1}{j\omega(1+0.2j\omega)(1+0.05j\omega)}$$

corner frequencies

$$\omega_{c1} = \frac{1}{0.2} = 5, \quad \omega_{c2} = \frac{1}{0.05} = 20$$

$$|G(j\omega)| = \frac{1}{\omega \sqrt{1+(0.2\omega)^2} \sqrt{1+(0.05\omega)^2}}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(0.2\omega) - \tan^{-1}(0.05\omega)$$

ω rad/s	2	3	5	10	12	15	20
$ G(j\omega) $	$\frac{46}{0.97} \approx 47$	0.28	0.13	0.04	0.02	0.01	≈ 0.008
$\angle G(j\omega)$	$\frac{117.5}{-124.4}$	-129.4	-149	-180	-188.3	-198.4	-210.9

ω	2	3	5	10	12	15	20
$G_R(j\omega)$	-0.231	-0.18	-0.11	-0.04	-0.01	-9.48×10^{-3}	-6.8×10^{-3}
$G_I(j\omega)$	-0.94	-0.22	-0.06	0	$\frac{+0.0002}{0.002}$	$\frac{3.15 \times 10^{-3}}{0.003}$	$\frac{1.1 \times 10^{-3}}{0.004}$

if G_m is given;

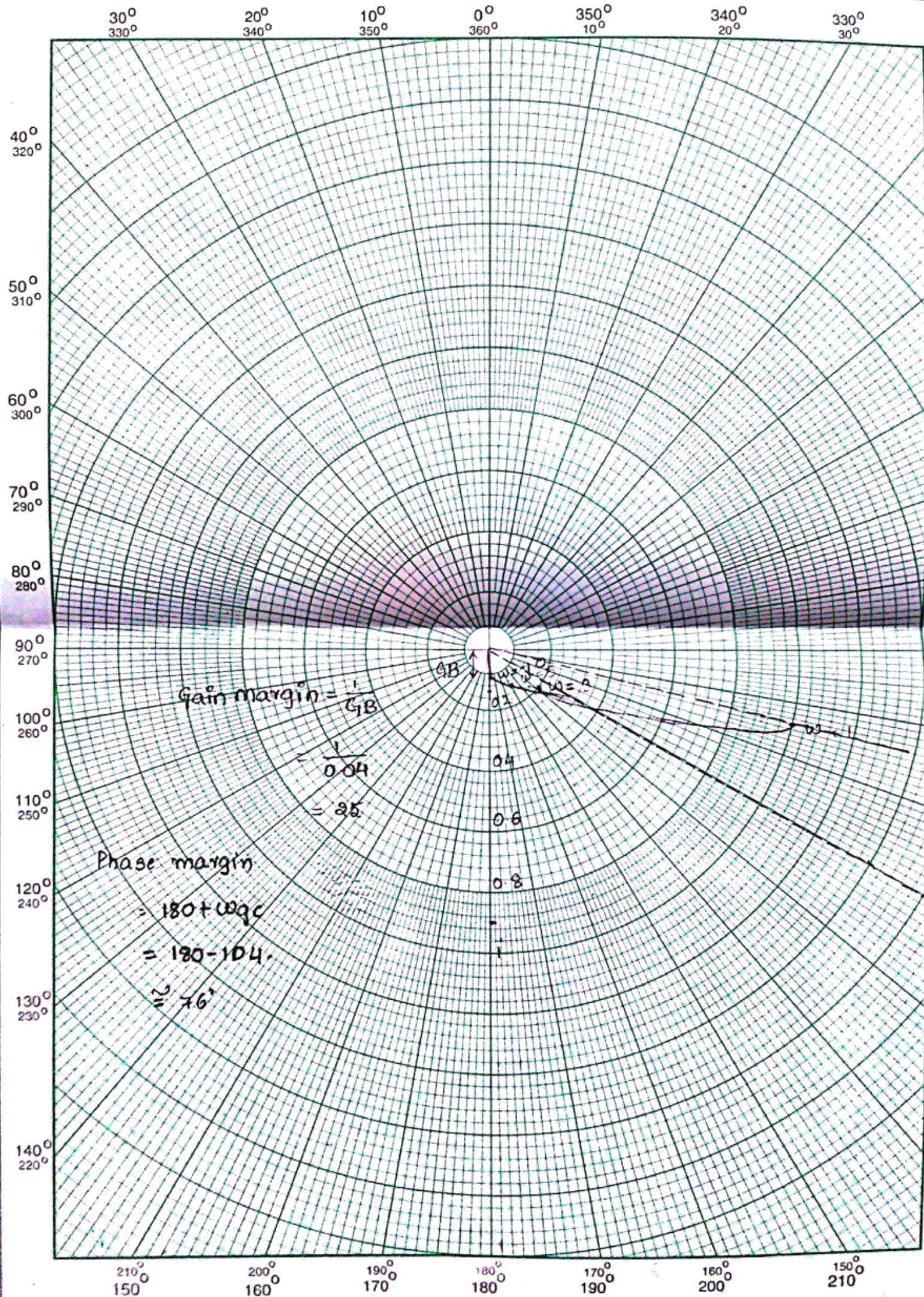
Given $G_m = 18 \text{ dB}$.

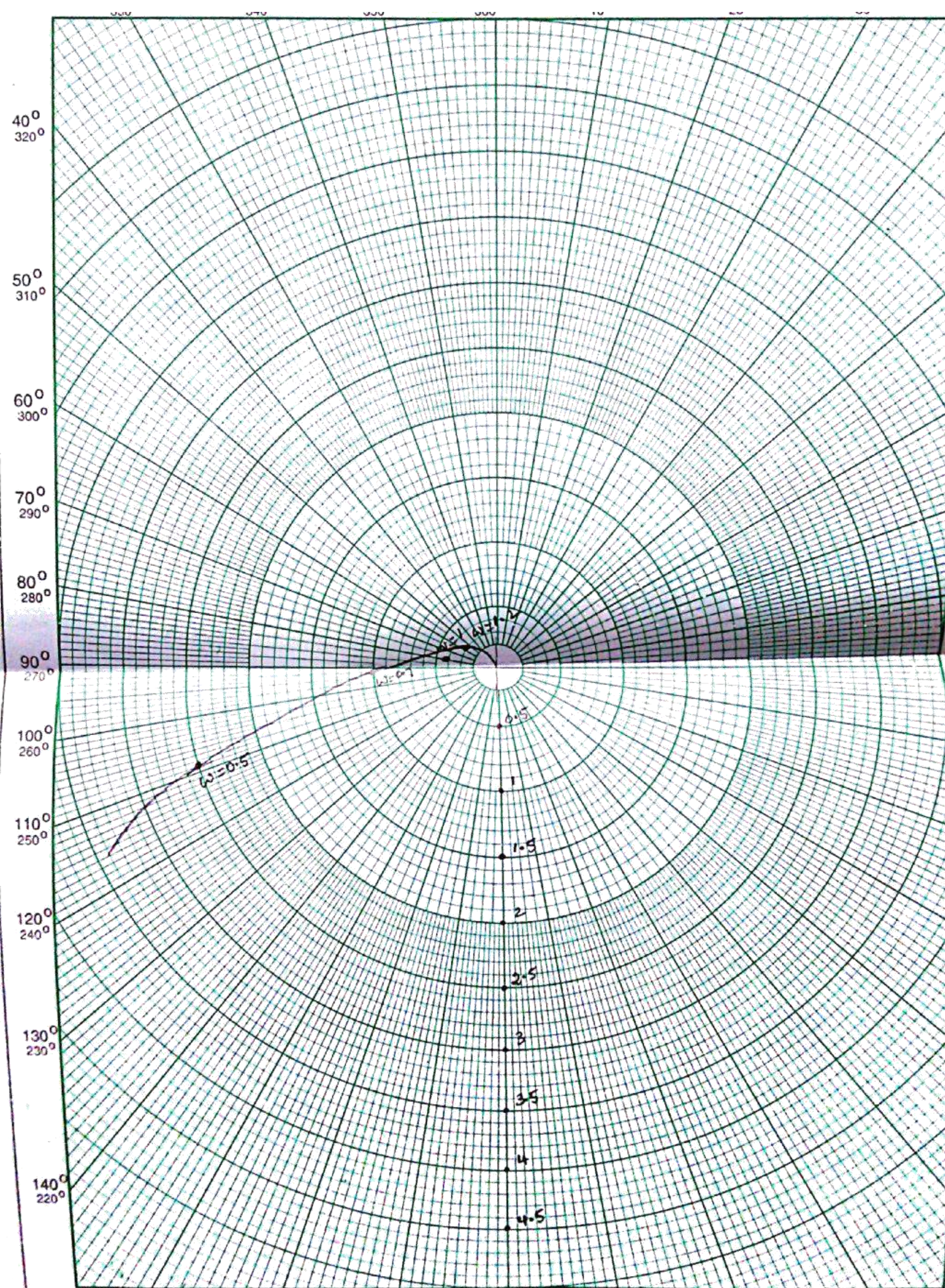
we got $G_B = 0.04$

let G_A be the gain at -180° for a G_m of 18dB

$$20 \log \frac{1}{G_A} = 18$$

$$G_A = 0.125$$





$$K = \frac{G_A}{G_B} = \frac{0.125}{0.04} = 3.125.$$

(0.9)

$$\text{let } x_1 = 0.04$$

$$20 \log \frac{1}{y_1} = 18$$

$$y_1 = 0.125$$

$$K = \frac{y_1}{x_1} = 3.125$$

If PM is given

$$PM = 60^\circ$$

let ϕ_{gc} be the phase of $G(j\omega)$ for a PM of 60°

$$60^\circ = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = -120^\circ.$$

let G_A be the magnitude of $G(j\omega)$ at the point where -120° line cuts the locus

$$K = \frac{1}{G_A} = \frac{1}{0.425} = 2.351.$$

Gain for actual PM, $x_2 = 1$

$$PM = 60$$

$y_2 =$ gain for specified PM

$$60^\circ = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = -120^\circ$$

y_2 be the gain for where -120° line cuts the locus

$$y_2 = 0.425$$

$$K = \frac{x_2}{y_2} = \frac{1}{0.425} = 2.35.$$

$$G(s) = \frac{1}{s^2(1+s)(1+2s)}$$

Sol:-

Replace 's' by 'j ω '

$$\Rightarrow G(j\omega) = \frac{1}{(j\omega)^2(1+j\omega)(1+2j\omega)}.$$

corner frequencies,

$$\omega_{c1} = \frac{1}{1} = 1, \omega_{c2} = \frac{1}{2} = 0.5 \Rightarrow \omega_{c1}$$

$$|G(j\omega)| = \frac{1}{\omega^2 \sqrt{1+(0.5\omega)^2} \sqrt{1+\omega^2}}$$

$$\angle G(j\omega) = -180^\circ - \tan^{-1}(0.5\omega) - \tan^{-1}(\omega)$$

ω rad/s	0.35	0.4	0.45	0.5	0.6	0.7	1
$ G(j\omega) $	7.58	5.69	4.39	3.47	2.88	1.57	0.63
$\angle G(j\omega)$	-119.21						

ω rad/sec	0.35	0.4	0.45	0.5	0.6	0.7	1
$G_R(j\omega)$	-5.12 6.31	-3.91	-3.05	-2.39			
$G_I(j\omega)$	-3.68	-2.21	-1.34	-0.77			

ω rad/s	0.35	0.4	0.45	0.5	0.6	0.7	1
$ G(j\omega) $	6.31	4.5	3.34	2.52	1.52	0.97	0.31
$\angle G(j\omega)$	-144.88	-150.46	-156.21	-161.56 -162.	-171.5	-179.45	-198.43

ω	0.2	0.4	0.5	0.7	1	1.2	1.4
$14(j\omega)$	-22.7	-4.53	-2.52	-0.97	-0.316	-0.170	-0.09
$4(j\omega)$	-213.11	-240.4	-251.5	-269.4	-288.5	-297.5	-304.8

ω	0.2	0.4	0.5	0.7	1	1.2	1.4
$G_R(\omega)$	19.01	2.23	0.79	0.01	0.09	0.07	0.05
$G_I(\omega)$	12.89	3.93	2.38	0.96	0.29	0.150	0.073

* Nyquist plots stability criterion :

So far we have discussed two methods of determining stability in time domain Analysis

1. RH criterion & Root-Locus method

- Nyquist stability criterion is another method which determines the stability of a closed loop system from open loop transfer function $G(s)H(s)$.

Features of Nyquist stability criterion :

1. Unlike RH method which provides information on absolute stability, Nyquist stability criterion gives information about both absolute & relative stability of system and degree of instability of unstable system & indicates how the system stability can be improved if necessary.
2. Nyquist plot can also be used to determine the stability of time delay system.
3. We can easily plot.
4. we can find frequency domain specification. i.e., M_r , w_r , B_w , Gain margin, Phase margin etc.
5. It is based on Cauchy's residue Theorem of complex variables which is referred to as "principle of arguments".

Nyquist stability criterion fundamentals :

1. Mapping :

A function $F(s)$ is said to be analytic

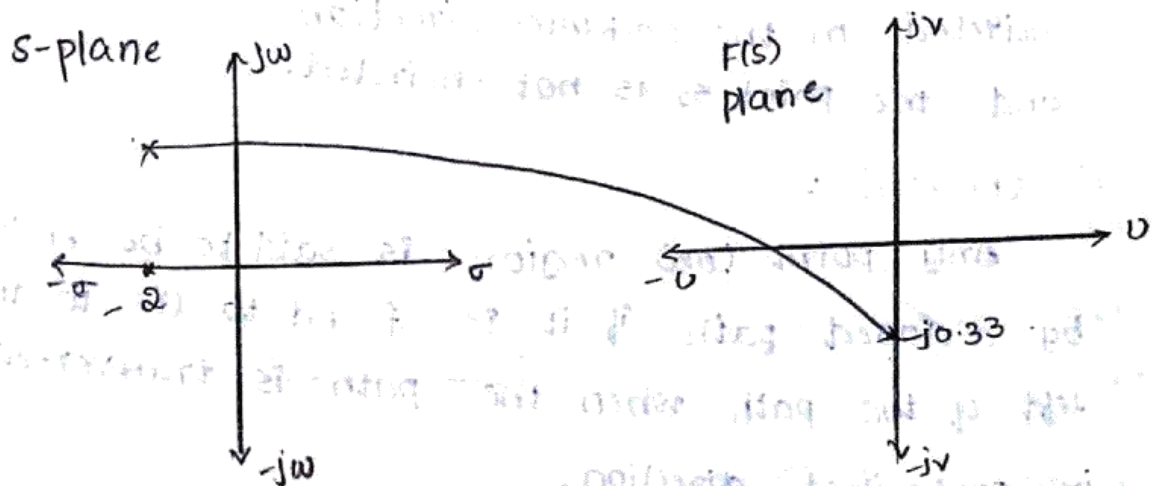
In the s-plane if the function and all its derivatives exist.

- The points in the s-plane where the function $F(s)$ or its derivatives does not exist are called singular points. Poles of $F(s)$ are singular points.
- Let us consider a function $F(s)$ which is analytic everywhere in the s-plane except at its singularity. Then there exists a point in the s-plane. For every point in s-plane at which $F(s)$ is adjacent. If we take a complex number $\sigma + j\omega$ in s-plane and substitute in $F(s)$ we get another complex number $u + jv$. This process is called mapping of $\sigma + j\omega$ into $u + jv$.

Ex: $F(s) = \frac{1}{s+2}$

Let $s = -2 + j0.33$

$\rightarrow F(s) = \frac{1}{-2 + j0.33 + 2} = \frac{1}{j0.33} = -j0.33$

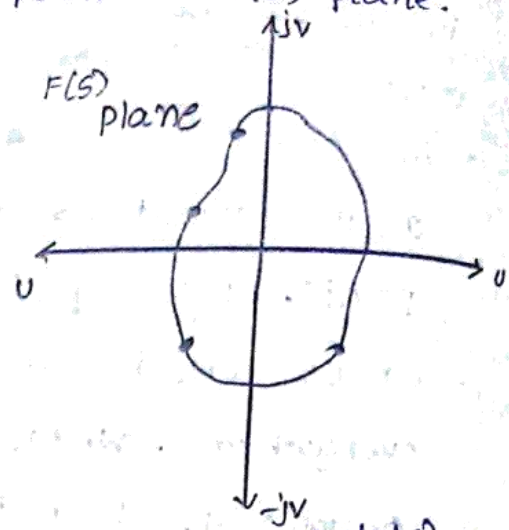
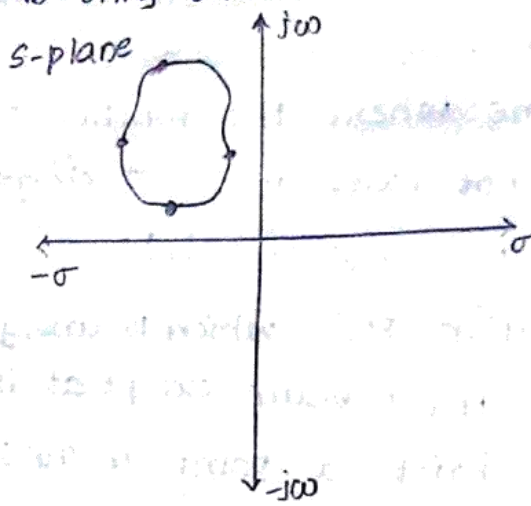


The point $s = -2 + j0.33$ in s-plane maps into $-j0.33$ in $F(s)$ -plane.

a Single valued Function :

It is defined as, 'for each point in s-plane, there

is only one corresponding point in $F(s)$ plane.



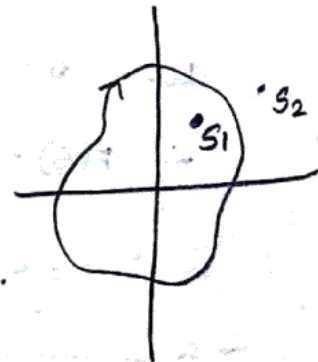
The direction of contour (collection of points) can be either clockwise or anticlockwise depending on the function $F(s)$.

Concept of encircled and enclosed.

a) encircled :

A point is said to be encircled by a closed path (closed contour) if it is found inside the path.

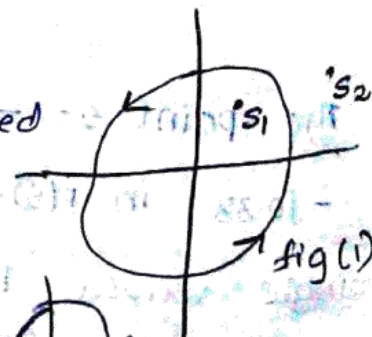
From the figure, the point s_1 is encircled in the clockwise direction and the point s_2 is not encircled.



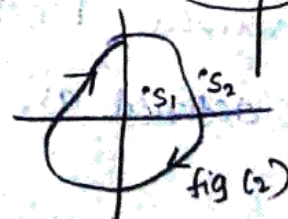
b) enclosed :

Any point (or) region is said to be enclosed by a closed path if it is found to lie to the left of the path when the path is traversed in prescribed direction.

From the figure, s_1 is enclosed by the closed path whereas s_2 is not.



In figure 2, s_2 is enclosed



Note :

The exact shape of the contour is not required but only the no. of encircle-ments of the origin of $F(s)$ plane is required.

Principle of

Principle of Argument is stated as follows .

Def :

The relation between the enclosure of poles and zeroes of $F(s)$ lying on right half s -plane by the s -plane contour & encirclements of the origin of $F(s)$ plane by the corresponding $F(s)$ contour is called principle of Argument.

Let $F(s)$ is a single valid function rational function and is analytic in a given region in s -plane except at some points.

Now if an arbitrary closed contour is chosen in s -plane, so that $F(s)$ is analytic at every point on the closed contour. Then, $F(s)$ plane ^{contour} is mapped in $F(s)$ plane will encircle the origin N -times in clockwise direction where N is the difference b/w no. of poles and no. of zeroes of $F(s)$ that are encircled by chosen closed contour in s -plane.

- Mathematically, it is expressed as $N = P - Z$ where N = no. of encirclements of the origin made by the contour in $F(s)$ plane.

P = no. of poles of $F(s)$ lying on right half of s -plane and enclosed by the s -plane closed contour.

Z = no. of zeroes of $F(s)$ lying on right half of s -plane

N can be +ve, -ve or zero. Based on the sign of N , the following conclusions can be made provided the arbitrary s -plane contour is chosen in clockwise direction.

1. If N is +ve, then the encirclement of origin of $F(s)$ plane will be in anticlockwise direction.
2. If N is -ve, then the encirclement of origin of $F(s)$ plane will be in clockwise direction.
3. If $N=0$, then there will be no encirclement of origin of $F(s)$ plane.

Nyquist stability criterion :

consider closed loop transfer function $\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$

characteristic eqn $1+G(s)H(s)=0$

Let $F(s) = 1+G(s)H(s)$

The loop transfer function can be expressed as

$$G(s)H(s) = \frac{k(s+z_1)(s+z_2) \dots (s+z_m)}{(s+p_1)(s+p_2) \dots (s+p_n)} \rightarrow \textcircled{1}$$

where $m \leq n$.

$$\Rightarrow F(s) = 1 + G(s)H(s)$$

$$= 1 + \frac{k(s+z_1)(s+z_2) \dots (s+z_m)}{(s+p_1)(s+p_2) \dots (s+p_n)}$$

$$F(s) = \frac{[(s+p_1)(s+p_2) \dots (s+p_n)] + [k(s+z_1)(s+z_2) \dots (s+z_m)]}{(s+p_1)(s+p_2) \dots (s+p_n)} \rightarrow \textcircled{2}$$

For the condition $F(s) = 1+G(s)H(s)=0$,

$$\Rightarrow [(s+p_1)(s+p_2) \dots (s+p_n)] + [k(s+z_1)(s+z_2) \dots (s+z_m)] = 0 \rightarrow \textcircled{3}$$

This is the characteristic equation of the system. For the stability of closed loop system, the zeroes of $F(s)$ should not lie on the right half of s -plane. They should lie in the left half s -plane i.e., open loop unstable system may lead to a closed loop stable operation.

- Let us choose an arbitrary contour in s -plane which encloses the right half zeroes and poles of $F(s)$.
- The principle of argument states the corresponding contour in $F(s)$ plane will encircle the origin of $F(s)$ N -times in anti-clockwise direction.
- For the stability of the system, the roots of characteristic equation and so the zeroes of $F(s)$ should not lie in right half s -plane. Hence for a stable system $z=0$.
- When $z=0$, $N = P \rightarrow \textcircled{4}$
- When $z \neq 0$, $N \neq P \rightarrow \textcircled{5}$

From Eqn-② we see that poles of $F(s)$ are also the poles of $G(s) \cdot H(s)$.

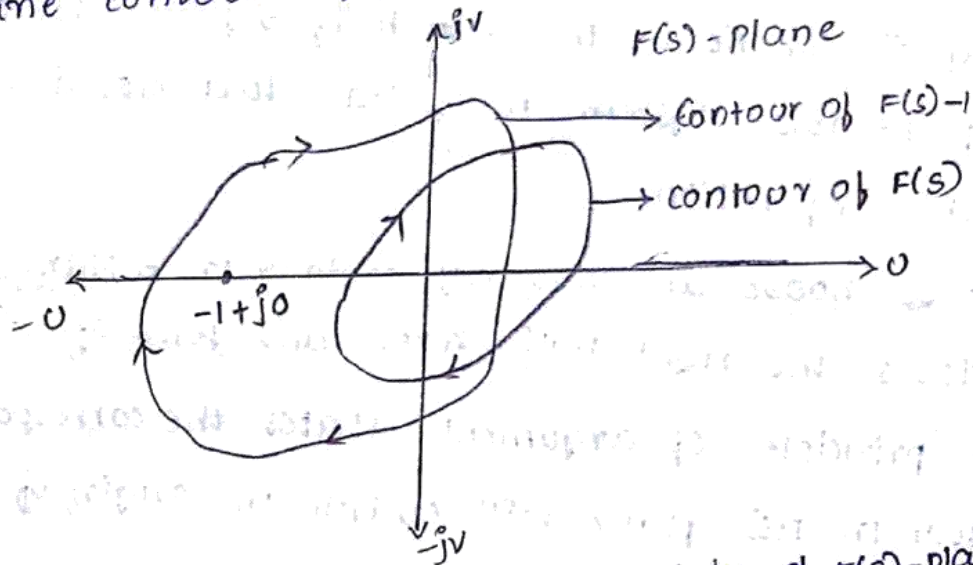
Hence for the stability of the system, the no. of poles of $G(s) \cdot H(s)$ lying on right half of s -plane should be equal to anti-clockwise encirclement of the origin of $F(s)$ plane.

If this condition is not satisfied, then the system is unstable.

The loop transfer function can be expressed as

$$G(s)H(s) = [1 + G(s)H(s)] - 1 = F(s) - 1 \rightarrow \textcircled{6}$$

From Eqn ⑥ it can be concluded that $F(s)$ -plane contour drawn w.r.t origin is same as $F(s)$ -plane contour of $[F(s)-1]$ drawn w.r.t $-1+j0$.



Therefore, the encirclement of origin of $F(s)$ -plane by $F(s)$ contour is equivalent to encirclement of the point $-1+j0$ by $G(s)H(s)$ contour.

The Nyquist stability criterion have been proposed based on this concept.

It is stated as follows [Nyquist stability criterion]

"If $G(s)H(s)$ contour in $G(s)H(s)$ plane corresponding to Nyquist contour in s -plane encircles the point $-1+j0$ in anti-clockwise direction as many times as the number of right half s -plane poles of $G(s)H(s)$, then the closed loop system is stable."

In determining the stability of linear control system using NSC, we come across 3 cases.

case (i):

"There is no encirclement of $-1+j0$ point."

= This implies that the system is stable if there are no poles of $G(s)H(s)$ in right half

s-plane. If there are poles on right half s-plane, then the system is unstable.

case (i) :

An Anti-clockwise encirclement of $-1+j0$ point.

- In this case, the system is stable if the no. of anticlockwise encirclements is same as no. of poles of $G(s) \cdot H(s)$ in the right half of s-plane.

- If the no. of encirclements is not equal to no. of poles of right half of s-plane, then the system is unstable :

case (ii) :

There is a clockwise encirclement of $-1+j0$ point.

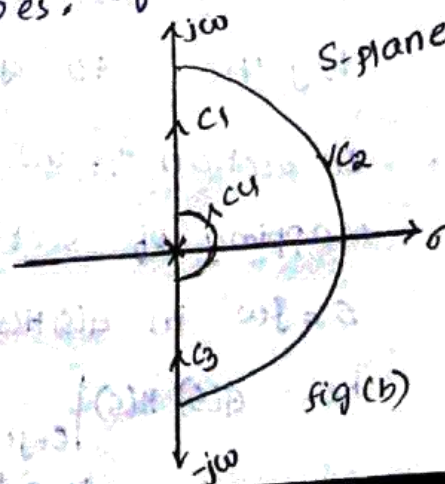
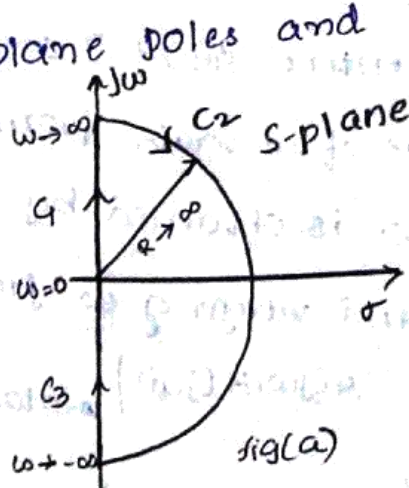
In this case, the system is always unstable.

Procedure for determining the stability using Nyquist stability criterion :

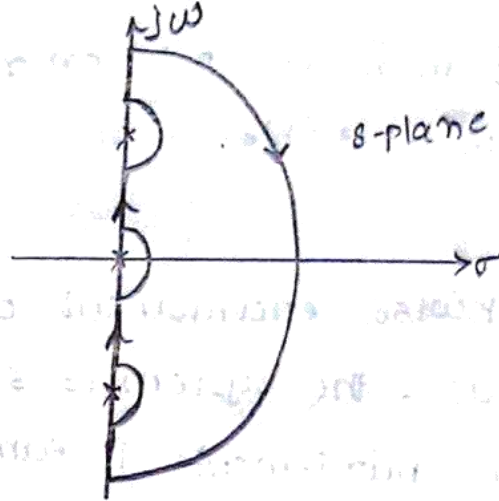
The following procedure can be followed to investigate the stability of closed loop system from the knowledge of open loop system using nyquist stability criterion .

1. Choose a Nyquist contour which encloses the entire right half s-plane except the singular points.

The Nyquist contour encloses all the right half s-plane poles and zeroes, of $G(s) H(s)$



fig(c)



fig(a):

Nyquist contour when there is no pole on imaginary axis

fig(b):

Nyquist contour when there is one pole at origin

fig(c):

Nyquist contour when there are poles on imaginary axis & at origin

2. The Nyquist contour should be mapped in $G(s)H(s)$ plane using the function $G(s)H(s)$ to determine encirclement $-1+j0$ point in $G(s)H(s)$ plane.

The Nyquist contour of fig(b) can be divided into 4 sections C_1, C_2, C_3 and C_4 .

Mapping of 4 sections in $G(s)H(s)$ plane can be carried section wise & then combined together to get entire $G(s)H(s)$ contour.

3. In section C_1 , the value of ω varies from 0 to ∞ . Mapping of section C_1 is obtained by equating $s = j\omega$ in $G(s)H(s)$ and varying ω from 0 to ∞ .

$$\text{i.e., } G(s)H(s) \Big|_{s=j\omega} = G(j\omega)H(j\omega) \Big|_{\omega=0 \text{ to } \infty}$$

4. Section c_2 has a semicircle of infinite radius. Therefore every point on c_2 has infinite magnitude but the argument varies from $+\pi/2$ to $-\pi/2$. Hence mapping of c_2 from s-plane to $G(s)H(s)$ plane can be obtained by equating $s = \lim_{R \rightarrow \infty} R e^{j\theta}$ in $G(s)H(s)$ plane and varying θ from $+\pi/2$ to $-\pi/2$.

5. In section c_3 , the value of ω varies $\rightarrow \infty$ from $-\infty$ to ∞ . Mapping of c_3 is obtained by equating $s = j\omega$ in $G(s)H(s)$ and varying ω from $-\infty$ to 0 .

$$G(s)H(s) \Big|_{\substack{s=j\omega \\ \omega=-\infty \text{ to } 0}} = G(j\omega)H(j\omega) \Big|_{\substack{s=j\omega \\ \omega=-\infty \text{ to } 0}}$$

6. Section c_4 has a semicircle of 0 radius. Therefore every point on semicircle has zero magnitude but the argument varies from $-\pi/2$ to $\pi/2$. Hence mapping of c_4 from s-plane to $G(s)H(s)$ plane can be obtained by equating $s = \lim_{R \rightarrow 0} R e^{j\theta}$ in $G(s)H(s)$ and varying θ from $-\pi/2$ to $\pi/2$.

1. Draw the Nyquist plot for the system whose open loop transfer function is $G(s)H(s) = \frac{k}{s(s+2)(s+10)}$.

Determine the range of k for which closed loop system is stable.

sol:- Step-1: Find time constant form for the given

$G(s)H(s)$.

Step-2: substitute $s = j\omega$ and find sinusoidal transfer function.

$$\Rightarrow G(s)H(s) = \frac{K}{2(10)\left[s\left(1+\frac{s}{2}\right)\left(1+\frac{s}{10}\right)\right]}$$

$$G(s)H(s) = \frac{K}{20s(1+s/2)(1+s/10)}$$

$$G(s)H(s) = \frac{0.05K}{s(1+0.5s)(1+0.1s)}$$

Replace s by $j\omega$.

$$G(j\omega)H(j\omega) = \frac{0.05K}{j\omega(1+0.5j\omega)(1+0.1j\omega)}$$

Note:

- mapping can be carried out for individual sections & all sections are combined

step-3: **G** mapping:

1. Find magnitude & phase expressions
2. Find -ve real axis intersection point.

method-1:

Equate imaginary part to 0

In $G(j\omega)H(j\omega)$. Then, we get ω_{pc}

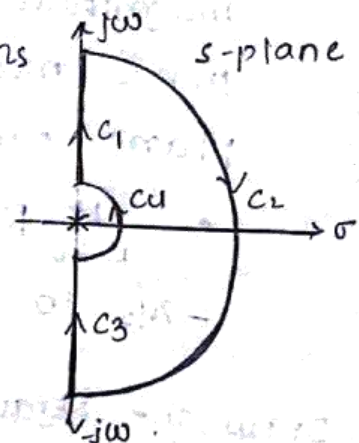
method-2:

Equate phase expression to -180° & solve, then

we get ω_{pc} .

- If we substitute this ω_{pc} in magnitude expression, then we get intersection point.

3. To find plot substitute $\omega=0$ and $\omega=\infty$ in magnitude & phase expressions & we get approximate plot. Another method is using



shortcut method

$$G(j\omega)H(j\omega) = \frac{0.05K}{j\omega(1 + 0.1j\omega + 0.5j\omega - 0.05\omega^2)}$$

$$= \frac{0.05K}{j\omega - 0.1\omega^2 - 0.5\omega^2 - 0.05\omega^3j}$$

$$= \frac{0.05K}{-0.6\omega^2 + j\omega(1 - 0.05\omega^2)}$$

When the locus of $G(j\omega)$ crosses real axis, the imaginary term will be zero and the corresponding frequency is ω_{pc} .

At $\omega = \omega_{pc}$

$$\omega(1 - 0.05\omega^2) = 0$$

$$\Rightarrow \omega_{pc}(1 - 0.05\omega_{pc}^2) = 0$$

$$1 - 0.05\omega_{pc}^2 = 0, \quad \omega_{pc} = 0$$

$$\omega_{pc}^2 \cdot 0.05 = 1$$

$$\omega_{pc}^2 = 20$$

$$\omega_{pc} = \pm \sqrt{20}$$

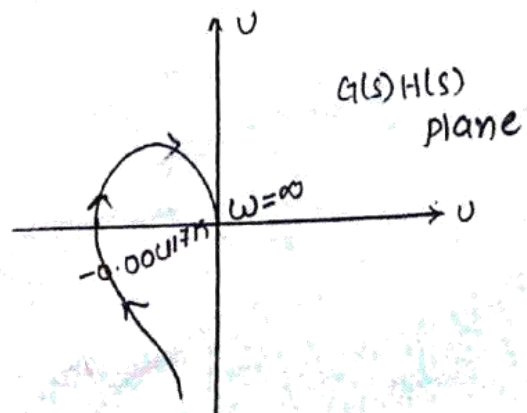
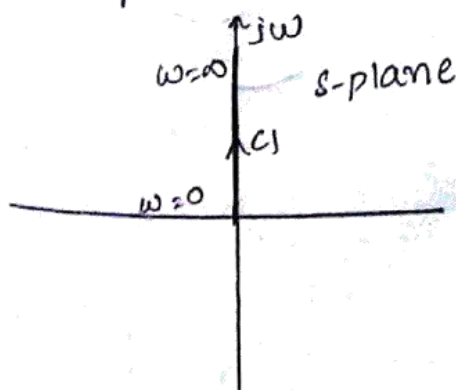
$$\omega_{pc} = 4.47$$

$$\omega_{pc} = 4.472 \text{ rad/sec}$$

subs. this value ω_{pc} in real part of $G(j\omega)H(j\omega)$

$$\Rightarrow \frac{0.05K}{-0.6\omega^2} = \frac{0.05K}{-0.6(4.47)^2} = -0.00417K$$

The open loop system is type-1, order-3



Step-4 : G_2 mapping :

1. This mapping can be done by replacing s with $\lim_{R \rightarrow \infty} Re^{j\theta}$ where θ is varying from $+\frac{\pi}{2}$ to $-\frac{\pi}{2}$

2. For this mapping $1 + ST$ is replaced by ST

$$G(s)H(s) = \frac{0.05K}{s(1+0.5s)(1+0.1s)}$$

Now $1+ST \in ST$

$$\Rightarrow G(s)H(s) = \frac{0.05K}{s(0.5s)(0.1s)}$$

$$\Rightarrow G(s)H(s) = \frac{+K}{s^3}$$

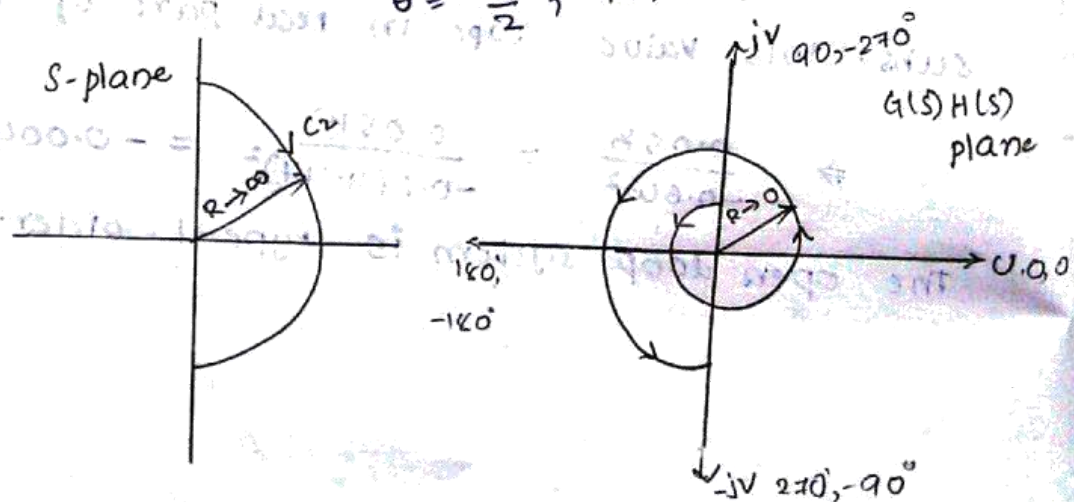
Replace s by $\lim_{R \rightarrow \infty} R e^{j\theta}$ $\Rightarrow \frac{k}{s^3} \Big|_{s = \lim_{R \rightarrow \infty} R e^{j\theta}}$

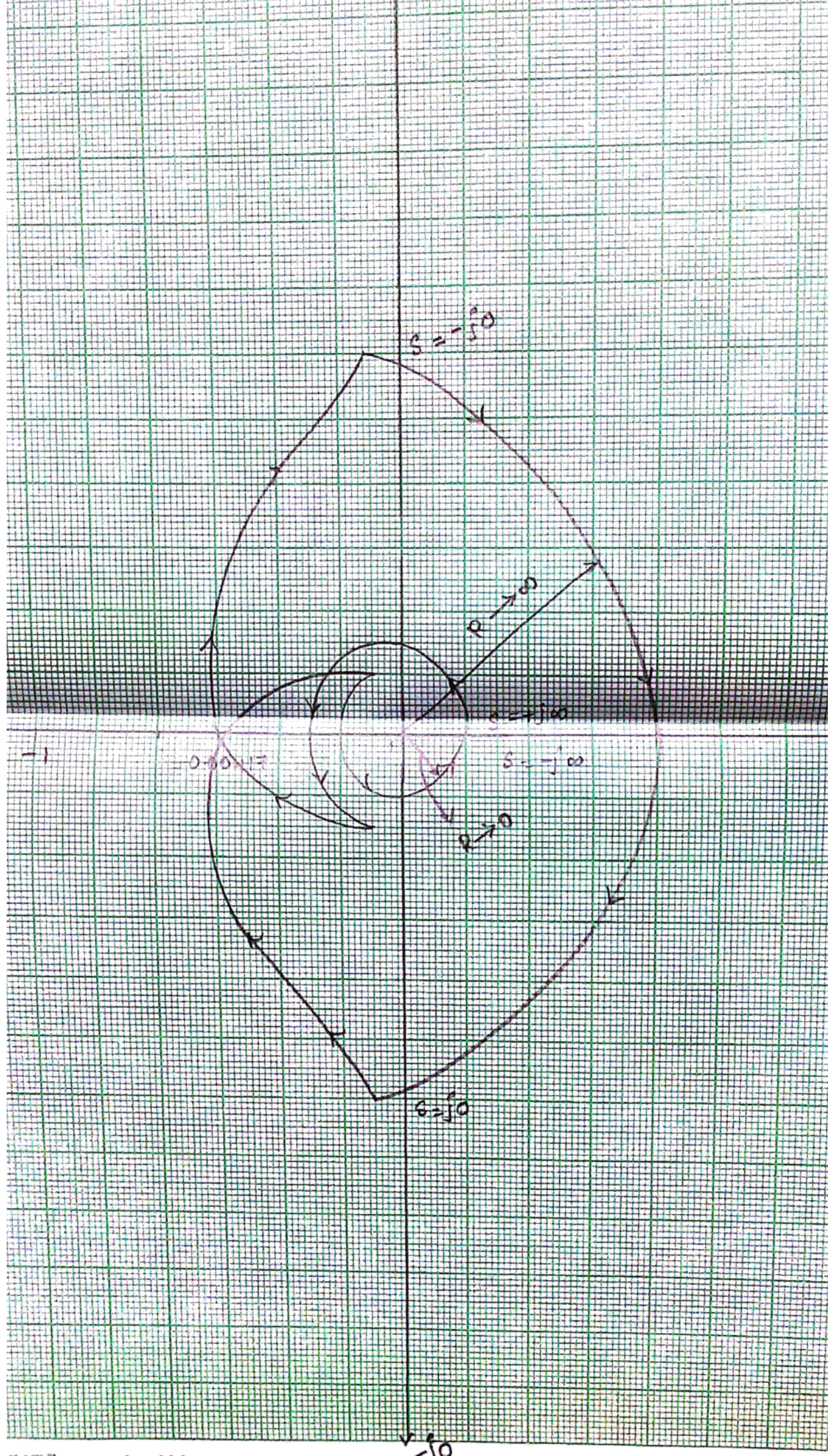
$$\Rightarrow G(s)H(s) = \frac{K}{s^3} = \frac{K}{1 + (Rej\theta)^3}$$

$$\Rightarrow \frac{k}{s^3} = 0. e^{-j3\theta} = G(s)H(s)$$

When $\theta = \frac{\pi}{2}$, $G(s)H(s) = 0 \cdot e^{-j\frac{3\pi}{2}}$

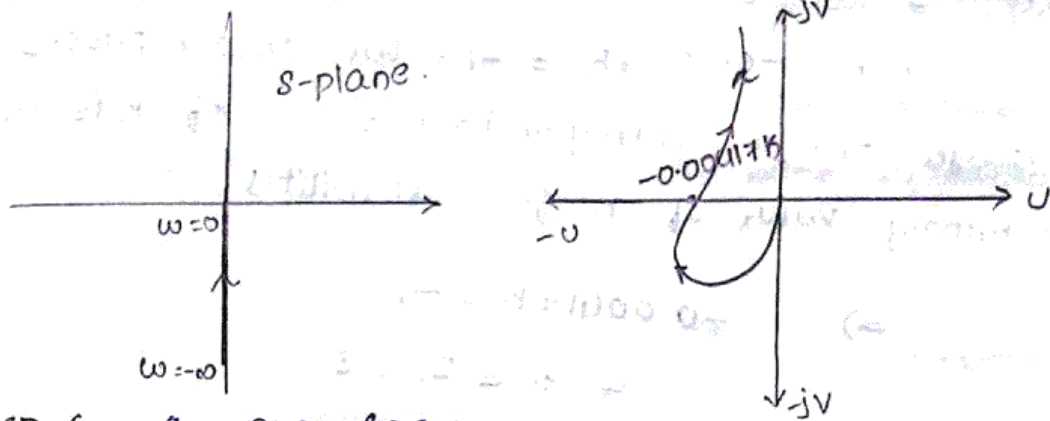
$$\theta = -\frac{\pi}{2}, \quad G(s)H(s) = 0.1 e^{j\frac{3\pi}{2}}$$





Step-5 : C_3 mapping :

It is the mirror image of C_1



Step-6 : C_4 mapping :

1. This mapping can be done by replacing s with

$\lim_{R \rightarrow 0} R e^{j\theta}$ where θ is varying from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.

2. For this mapping $1+sT$ is replaced with 1.

$$G(s)H(s) = \frac{0.05K}{s(1+0.5s)(1+0.1s)}$$

$$\text{Now } 1+sT \approx 1$$

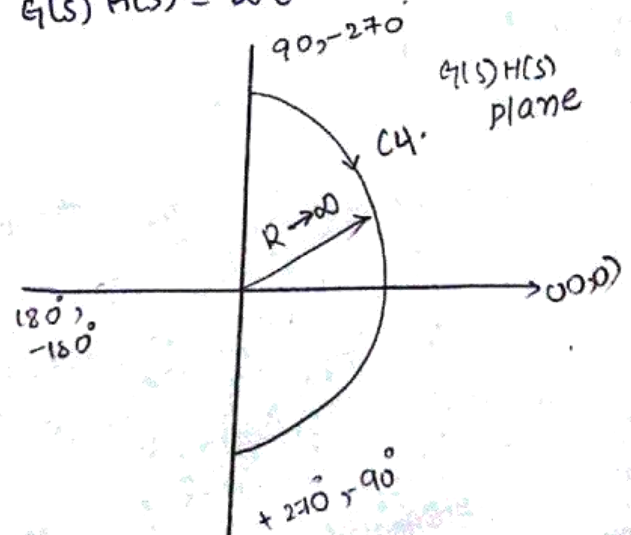
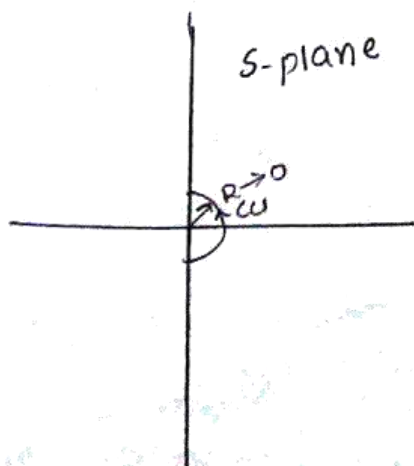
$$\Rightarrow G(s)H(s) = \frac{0.05K}{s}$$

Replace s by $\lim_{R \rightarrow 0} R e^{j\theta}$

$$\Rightarrow G(s)H(s) = \frac{0.05K}{\lim_{R \rightarrow 0} R e^{j\theta}} = \infty e^{-j\theta}$$

$$\text{When } \theta = -\frac{\pi}{2} \Rightarrow G(s)H(s) = \infty e^{j\frac{\pi}{2}}$$

$$\theta = +\frac{\pi}{2} \Rightarrow G(s)H(s) = \infty e^{-j\frac{\pi}{2}}$$



Stability Analysis:

When $-0.00417K = -1$, the contour passes through $-1 + j0$. corresponding value of K is the limiting value of K for stability

$$\Rightarrow -0.00417K = -1$$

$$\Rightarrow K = 239.8$$

$$K \approx 240$$

$\therefore$ The value of K for stability is $0 < K < 240$